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List of Acronyms

AD	: Applicable document
API	: Application Programming Interface
ARM	: Atmospheric Radiation Measurement
ASCAT	: Advanced Scatterometer
ATBD	: Algorithmic Theoretical Basis Document
BfG	: German Federal Institute of Hydrology
CAL/VAL	: Calibration/Validation
CEMS	: Copernicus Emergency Management Service
CEOS	: Committee on Earth Observation Satellites
CHIRPS	: Climate Hazards Group InfraRed Precipitation with Station data
CGLOPS	: Global Land Component of the Copernicus Land Service
CLMS	: Copernicus Land Monitoring Service
COG	: Cloud Optimized Geotiff
CUR	: Current validation period (i.e. 2023)
DWD	: Deutscher Wetterdienst
ECMWF	: European Centre for Medium-Range Weather Forecasts
EDO	: European Drought Observatory
EU	: European Union
EUMETSAT	: European Organisation for the Exploitation of Meteorological Satellites
IMERG	: Integrated Multi-satellitE Retrievals for Global Precipitation Measurement
fAPAR	: Fraction of absorbed photosynthetically active radiation
FMI	: Finnish Meteorological Institute
FR Aqui	: Institute of Agricultural Research
FWC	: Framework Contract
GBOV	: Copernicus Ground-Based Observations for Validation
GCOS	: Global Climate Observing System
GDO	: Global Drought Observatory of CEMS
GLDAS	: Global Land Data Assimilation System
GMAO	: NASA's Global Modeling and Assimilation Office
GPM	: Global Precipitation Measurement
H SAF	: Satellite Application Facility on Support to Operational Hydrology and Water Management
ICWRGC	: International Center for Water Resources and Global Change
INSPIRE	: Infrastructure for spatial information in Europe
ISMN	: International Soil Moisture Network
JRC	: Joint Research Centre of the European Union
KMD	: Kenya Meteorological department
LPV	: Land Product Validation
MERRA-2	: Modern Era Retrospective Analysis for Research and Applications, version 2

NASA	: National Aeronautics and Space Administration
NetCDF	: Network Common Data Form file format
NF	: News Footage
Noah-LSM	: Noah Land Surface Model
NRT	: Near Real Time
PQAD	: Product Quality Assurance Document
PUM	: Product User Manual
rho	: Spearman correlation coefficient (rho)
REF	: Reference validation period (i.e. 2007-2018)
RHEMEDUS	: In-Situ Network of the Centro Hispano Luso de Invest. Agrarias, Salamanca, Spain
RMSD	: Root Mean Square Difference
RSMN	: In-Situ Network of the National Meteorological Administration, Romania
SALLJ	: South American Low-Level Jet
SCAN	: Soil Climate Analysis Network
SMAP	: Soil Moisture Active Passive
SNOTEL	: In-Situ Network of the Snow Telemetry
SQAD	: Service Quality Assurance Document
SQE	: Scientific Quality Evaluation
SPI (-12)	: Standardized Precipitation Index (on the 12 months' time scale)
SSF	: Surface State Flag
SSM	: Surface Soil Moisture
STEMS	: Institute of Sciences and Technologies for Sustainable Energy and Mobility
SWI	: Soil Water Index
SWI10	: Soil Water Index 10-daily average
SWI-TS	: Soil Water Index Time Series
SYNOP	: Surface Synoptic Observations
TAHMO	: Trans-African Hydro-Meteorological Observatory
TERENO	: In-Situ Network of the Terrestrial Environmental Observatories
TxSON	: Texas Soil Observation Network
UN	: United Nations
USCRN	: U.S. Climate Reference Network
WGS84	: World Geodetic System 1984

EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. The production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time series.

This document describes the Scientific Quality Monitoring of the Soil Water Index (SWI) Version 3 (V3) products for the year 2024. In accordance with the System Quality Assurance Document (SQAD) [CGLOPS1_SQAD], the soil moisture quality monitoring within CLMS has been transformed and further automated, the product quality analysis results are compiled as an annual update of the Quality Assessment Report (QAR).

The aim of this annual global analysis is to assess the performance of the near real time (NRT) SWI V3 product in the period from 2024-01-01 until 2024-12-31, relative to the performance of the SWI archive available from the CLMS of the previous years, starting in 2007. The validation presented in this report is performed according to the procedures defined in the Product Quality Assurance Document [CGLOPS1_PQAD]. The regional analyses focus on drought in Central Europe, and on heavy rainfalls leading to flood events in East Africa and southern Brazil.

For the global analysis, the SWI and Surface State Flag (SSF) is compared to the output of a land surface model as well as in-situ data. The used model is GLDAS-Noah V2.1 (Global Land Data Assimilation System), whereas the in-situ data is taken from the International Soil Moisture Network (ISMN). The SWI is then compared to different soil moisture layers of these reference datasets, whereas the SSF is assessed using the first soil temperature layer of GLDAS. The compliance matrices show very similar results to those from previous yearly evaluations, documenting a stable performance of the SWI product. Also, the spatial patterns of agreement with GLDAS for SWI and SSF are widely consistent with the previous analysis, with a weak performance over tropical rainforests, subarctic areas, and deserts, and a good performance over Europe, subtropical Africa, India, Central and Eastern Asia, non-arid Australia and the (non-polar) Americas. Overall, the quality of the SWI and SSF products appears unchanged, with some regional differences in both positive and negative direction.

In the in-situ analysis, overall scores are very similar to the results from the previous years. As the magnitude of these changes is little and in the range of previous years, a year-to-year fluctuation can be attested, with no significant long-term trend apparent.

In the regional analyses over Central Europe, southern Brazil, and East Africa, overall good agreements are observed in 2024 between SWI anomalies compared to vegetation, temperature and precipitation datasets. Hydrological anomaly events such as droughts (Hungary and Serbia) or floods (Kenya, Tanzania and Brazil) described by news footage correspond spatiotemporally with significant SWI anomalies. The exceptional drought in summer 2024 over Hungary and northern Serbia is captured by the fAPAR index, the Standardized Precipitation Index (SPI), and the SWI anomalies. Also, the spatial patterns stemming from heavy rainfalls observed over East Africa and south of Brazil in the external anomaly datasets on SPI, 10-meter air temperature, mean precipitation, 500-hPa height and winds, water vapor and monthly rainfall are well replicated by the SWI anomaly data.

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

The document presents the results of the annual scientific quality monitoring of the operational SWI V3.0 (SWIv3) product. This product provides global daily soil moisture images and time series data from January 2007 ongoing.

The quality evaluation is performed on SWI data for the year from 2024-01-01 to 2024-12-31 with a reference period from 2007-01-01 to 2023-12-31.

The objective is to check that the operational products keep the same level of quality in the period under study than the fully validated products.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the users' requirements, and the expected performance
- Chapter 3 reviews the results of previous validation reports
- Chapter 4 describes the methodology for quality assessment, the metrics and the criteria of evaluation
- Chapter 5 presents the results of the analysis
- Chapter 6 summarizes the main conclusions of the study
- Chapter 7 makes recommendations based upon the results

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

1.3.2 Input

Document ID	Descriptor
CGLOPS1_PQAD	Product Quality Assurance Document of the Global Land Service.
CGLOPS1_SQAD	System Quality Assurance Document of the Global Land Service.
GIOGL1_ATBD_SWIV3	Algorithm Theoretical Basis Document of the SWI V3.0 product.

Available at: https://land.copernicus.eu/en/technical-library

1.3.3 Output

Document ID	Descriptor
CGLOPS1_PUM_SWIV3-SWI10-SWI-TS	Product User Manual summarizing all information about the SWI V3.0, SWI10 and SWI-TS products.

Available at https://land.copernicus.eu/en/technical-library

1.3.4 External documents

Document ID	Descriptor
GCOS-245	The 2022 GCOS ECVs Requirements https://library.wmo.int/records/item/58111-the-2022-gcos-ecvs-requirements-gcos-245?offset=1
SMAP Handbook	Mapping Soil Moisture and Freeze/Thaw from Space. NASA. https://smap.jpl.nasa.gov/system/internal_resources/details/original/178_SMAP_Handbook_FINAL_1_JULY_2014_Web.pdf

2 USERS REQUIREMENTS

According to the applicable document [AD1], the user's requirements relevant for SWI1v3 are described below.

2.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

SWI – Soil Water Index	
The surface soil water index provides daily information about moisture conditions in different soil depths. Soil moisture is a very heterogeneous variable and varies on small scales with soil properties and drainage patterns. Satellite measurements integrate over relative large-scale areas, with the presence of vegetation adding complexity to the interpretation. The SWI quantifies the amount of water (m^3/m^3) in soil layers at various depths and is derived from the SSM using a water balance model. The freeze and thaw status of soil is provided.	
Geometric Properties:	
spatial resolutions	3-10km/1km
geolocation precision	Better than 0.5 pixels
coordinate position	Centre of the pixel
geodetical datum	WGS84
geographic projection	regular latitude/longitude grid
Geographic Coverage:	Global/EU
Time span of output product:	SWI: Daily SWI10: 10-day period SWI TS (timeseries): 6 months
Timeliness of output delivery:	SWI: 12hrs SWI-10: within 3 days after synthesis period
Uncertainty	Threshold: 0.1 m^3/m^3 Goal: 0.04 m^3/m^3
Stability	Threshold: 5% Goal: 3%

2.2 FURTHER REQUIREMENTS

2.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

2.2.2 Data Structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.
Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

2.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets will be available in NETCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the CA during the FWC.

2.2.4 Uncertainties and Validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS-245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Level 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component when biophysical parameters are available.

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

2.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

Land Surface Temperature (LST) and SWI can be based on geostationary and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

2.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination.
The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic

3 SUMMARY OF THE PREVIOUS SWI QUALITY MONITORING FOR 2023

In the continuous evaluation process of the CLMS, the scientific quality monitoring of the SWI V3.0 2023 products was carried out in the last year's scientific quality monitoring cycle. A summary of the results related to the user requirements is presented below.

In the 2023 VR, the SWI V3.0 product was compared to data from the Global Land Data Assimilation System (GLDAS) as well as in-situ data from the International Soil Moisture Network (ISMN). Table 1 shows the compliance matrices for 2023 product compared to products for the reference time period between 2007 and 2022. It can be seen that, both GLDAS and ISMN validations show very good results for the threshold SWI value of 0.10 m3/m3 for all analysed T-values (T = 1 / 20 / 100). As for the goal value of SWI of 0.04 m^3/m^3, an increase compared to the reference period across all T-values can be seen.

Table 1: VR 2023: Combined compliance matrix for the comparison with GLDAS Noah and ISMN. Percentages of grid points that fulfil each requirement. Left, the reference (REF) period (2007-01-01 until 2022-12-31) / right, the validation period (2023-01-01 until 2023-12-31) (CUR)

Variable	CGLOPS Threshold				CGLOPS Goal			
	SWI 0.10 m^3/m^3				**SWI 0.04 m^3/m^3**			
	GLDAS		ISMN		GLDAS		ISMN	
[%]	*ref*	***cur***	*ref*	***cur***	*ref*	***cur***	*ref*	***cur***
SWI T=1	*99*	***98***	*82*	***89***	*30*	***36***	*19*	***21***
SWI T=20	*99*	***97***	*87*	***93***	*24*	***30***	*29*	***30***
SWI T=100	*98*	***94***	*81*	***91***	*37*	***39***	*29*	***36***

4 SCIENTIFIC QUALITY EVALUATION METHOD

4.1 GLOBAL ANALYSIS

The aim of the global analysis in this report is to assess the performance of the SWI V3 NRT product in the year 2024, relative to the performance of the SWI archive available from the Copernicus Land Monitoring Service of the previous years (starting in 2007). The validation presented in this report is performed in accordance with the procedures defined in the Product Quality Assurance Document [CGLOPS1_PQAD], but are also shortly summarized in the following.

For the SWI validation, the following metrics are used:

- Pearson's correlation coefficient (rho) with a p value < 0.05
- Root mean square difference (RMSD)

The reference datasets (GLDAS and ISMN situ data) are temporally matched to the SWI observation timestamp using nearest neighbour matching. This is necessary since the SWI is a daily dataset, whereas the reference datasets have multiple observations per day.

Since the SWI data is provided in percentage of relative saturation (i.e. %) while the reference products are provided in m^3/m^3 (ranging commonly from 0.0 – 0.5 m^3/m^3), data scaling is necessary prior to the evaluation of difference metrics such as RMSD. Following leading experts and current literature scaling into the in-situ data space is done (Brocca et al., 2011, Montzka et al, 2020).

In order to report here the targeted metrics in m^3/m^3, "mean-standard-deviation scaling" (also known as z-normalisation) is used to scale the SWI data to the reference products (Lima and Souza, 2023). With this method, the SWI data is scaled to have the same mean value and standard deviation as the reference dataset. This approach was chosen because it is widely adopted by the community and because the extent of the scaled values is less affected by the presence of outliers compared to "minimum-maximum scaling", which was used in previous VR reports. Moreover, through the mean-standard-deviation scaling, no statistical bias remains between CLMS and reference time series, and the obtained RMSD values are equivalent to the unbiased Root Mean Square Error (ubRMSE), a metric regularly used within good practices established by the CEOS LPV (Montzka et al, 2020). Contrary to the VR reports prior to 2023, the bias analysis has been dropped from the annual validation, also recognising the large congruence of RMSD- and bias- patterns hitherto, and the repetitive insights drawn therefrom. And again, this streamlining allows applying the community-standard of mean-standard-deviation scaling (which removes the statistical bias from the data in the process).

Before calculating the metrics, observations are removed for which the Surface State Flag (SSF) shows frozen conditions. If less than 10 observations are available for a grid point, no metrics are

calculated. It should be noted that a changing sample size due to ISMN flag updates (e.g. from updated frozen flagging, saturated plateaus, snow or excessive precipitation) – and related shifts in mean and standard deviation – can cause different appearance of the scaled SWI data in the different annual VR reports, accordingly (or even alter the decision if the location falls below the threshold of 10 matching observations).

For the Pearson correlation coefficient rho, the 95% confidence intervals are calculated and used to determine if there are significant differences between the reference period and the current validation period (January-December 2024). Confidence intervals were estimated using the Fisher z-transformation (Fisher, 1915), so that for each rho-value a confidence band is found. Two rho-values are only considered significantly different when their confidence bands do not overlap.

In addition to the global rho-maps, also seasonal/quarterly time series of average rho-values over specific regions are calculated. The regions were chosen to well represent different climate zones and biomes, comprising subcontinent-scale areas (Continental USA, eastern Africa, Australia) as well as state-scale areas (Patagonia, Italy, California).

For this, the temporally and spatially matched values from the SWI time series and the reference time series of all points in a region per season/quarter are used to calculate a rho-value for this region over the three months periods. The chosen regions are of sub-continental- or country- scale and are shown in Figure 1. The rho-values are calculated for the reference period and the current validation period. For both periods, only data from pixels are used where each period had valid observations. However, this does not ensure the same number of values per year, since the freeze/thaw masking is different from year to year. For the reference dataset, the standard deviation of the quarterly rho-values is computed per chosen region, and added to the plots to give insight on the relevance of the difference of current to reference data.

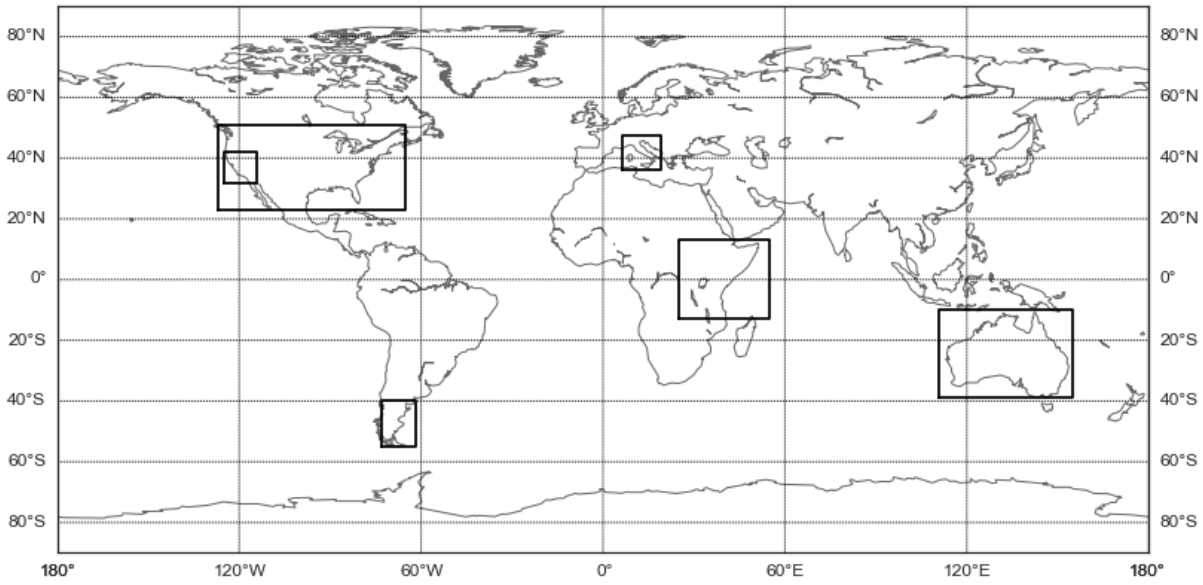


Figure 1: Map showing the regions used to define the regions of special interest for the plots of Pearson correlation rho over time.

For the SSF validation, the following metrics are used:

- Percentage of correctly classified observations (*percent_valid*)
- Period of consecutively misclassified days (mean, median) (*wrong_days*)

Classification of the SSF values (Table 2) into the binary form of "correct-or-incorrect" is done using the classification scheme outlined in Table 3. New in this year's VR, the soil temperature is taken as the daily *minimum* temperature from the first layer of the used land surface model (see Section 4.3). The daily minimum value is selected to reduce the impact of the daily temperature cycle. The *percent_valid* metric is then calculated by calculating the percentage of correctly classified observations as a fraction of all observations. The *wrong_days metric* is calculated by finding consecutive periods of wrong classification and measuring their length in days. Since each grid point can have multiple such periods the mean and median length of these periods is stored. During the calculation of the *wrong_days*-metric, every year is taken by itself to make the results comparable to the current validation period (that spans one single year). Please note that this has a considerable effect on the obtained results, and different values compared to previous years' analysis are expected.

Table 2: Possible values of the SSF and their meaning

SSF value	Detected surface state
255	Could not be determined
0	Unknown
1	Unfrozen
2	Frozen
3	Temporary water on the surface

Table 3: Classification scheme

SSF value	Soil temperature	Classification
0, 1, 3	>= 0°C	Correct
2	< 0°C	Correct
0, 1, 3	< 0°C	Incorrect
2	>= 0°C	Incorrect

The SSF analysis bears the caveat that soil temperature is only a rough proxy for the freeze/thaw state of the soil. However, it is to our knowledge the only possibility for SSF validation at the moment. Several studies working with freeze/thaw data from active (Naeimi et al., 2012) and passive (Youngwook et al., 2011) satellites as well as the Soil Moisture Active-Passive (SMAP) sensor freeze/thaw product [SMAP Handbook] rely on temperature observations for such validation purposes. Furthermore, since the annual VR reports are due three months after the investigated SWI data has been produced, we can only use datasets that are updated in a timely manner. This adds to the complexity of finding an alternative dataset to temperature for SSF validation. Unfortunately, the availability and reliability of temperature data in the ISMN soil temperature dataset remain challenging to use, as we find numerous and erratic data-gaps and -flaws (see Section 4.4). Consequently, we limit the evaluation of the SSF to comparison against the global GLDAS model data.

4.2 REGIONAL ANALYSIS

For the regional analysis, we investigate in-depth the hydrological dynamics over regions affected by extreme conditions. For 2024, southern Brazil, Central Europe, and eastern Africa are in the focus.

To examine the hydrology of these regions, we build the monthly anomalies from the decadal CLMS SWI10 product (that already aggregated the SWI signal by a 10-day sampling), allowing efficient and robust computation of monthly means. In more detail, we calculate the anomalies as follows:

1. Building the monthly means for each month of the full SWI data period 2007-2024.
2. The monthly means of the reference period (2007-2023) are averaged again for each calendar month, yielding per pixel 12 values, and building the reference.
3. Again, per calendar month, subtracting the reference monthly means from the 2024-monthly-means.

The subsurface SWI (T=1) monthly anomalies are then compared qualitatively against the Standardized Precipitation Index (SPI) with an accumulation period of 12 month (SPI-12), fAPAR anomalies (specific for the drought event in Section 5.2.3), 10-meter air temperature anomaly, precipitation water vapour anomaly, anomalous height at 500-hPa and winds, mean precipitation, and rainfall anomaly. A detailed descriptions and references of the products are provided in Section 4.5.

4.3 Model Reference Products

Modelled reference products to which the SWI product is compared against are taken from the GLDAS (Global Land Data Assimilation System) Noah-LSM (Land Surface Model), version 2.1. The GLDAS Noah Land Surface Model (Rodell et al., 2004, Beaudoing et al., 2020) is produced by NASA. It includes four soil layers with the bottom depth at 0.1, 0.4, 1.0, and 2.0 m respectively, and simulates over 30 land surface parameters in total. Data is available with a temporal sampling of three hours on a global 0.25° grid. The model soil moisture data was converted from kg/m² to m^3/m^3 by using the formula SM [m^3/m^3] = SM [kg/m^2] * 0.001 * 1/d, where d is the thickness of the soil layer in meters. The factor 0.001 is due to the assumption that 1 kg of water represents 1000 cm^3, which is 0.001m^3.

The GLDAS model dataset is chosen for the validation because it provides consistent long-term global data, and is updated regularly at relatively small latency. The consistency of the dataset also makes it likely that future VR exercises can use them as well. This would have been less likely if we had chosen e.g. soil moisture products derived from other remote sensing platforms that are exposed to risks lying in single systems, e.g. which are vulnerable to satellite failure.

The soil moisture component of the model is, of course, not free of errors, but there are comprehensive studies on the global quality of the modelled soil moisture. Dorigo et al. (2010) investigated the quality of the GLDAS and other models with the triple collocation method using remotely sensed soil moisture datasets as the reference datasets and found consistent performances among global modelled soil moisture datasets. Chen et al. (2013) compared the GLDAS model output against in-situ stations

on the Tibetan plateau and found that the model underestimates surface soil moisture (0-5 cm) but was in good agreement with a deeper layer (10 – 40 cm). Correlations of 0.69 (0 – 5 cm) and 0.76 (10 – 40 cm) were found. Because of these results suggesting a high agreement of the models, and following review board's recommendation, the GLDAS data is used for the SWI evaluation, as it also features a relatively high resolution (0.25°) for a global model dataset, which is acceptable in respect to the SWI's resolution of 0.1°.

For the VR 2024, the GLDAS-Noah Version 2.1 was used to cover the full evaluation period (as in the former VR reports 2017-2023).

4.4 IN-SITU REFERENCE PRODUCTS

For in-situ validation purposes, the project consortium can access the comprehensive database of the ISMN (International Soil Moisture Network; Dorigo et al., 2011, Dorigo et al, 2021). It is an international cooperation to establish and maintain a global in-situ Soil Moisture (SM) database, and is recommended by CEOS' Soil Moisture Product Validation Good Practices Protocol (Montzka et al., 2020).

This database is an essential means of the geoscientific community for validating and improving global satellite observations and land surface models. As of March 2024, filtered for the period 2007 to 2024, the ISMN contains data of 86 networks and more than 4000 stations located all around the globe, with highest density over Europe and North America, and most datasets developed during the last two decades. The database is rapidly expanding, which means that both the number of stations and the period covered by the existing stations are still growing. Hence, it will become an increasingly important resource for validating and improving satellite-derived soil moisture products and studying climate related trends. As the ISMN is animated by the scientific community itself, potential networks are invited to enrich the collection by sharing its in-situ soil moisture data.

The ISMN employs an automated quality control system for its SM measurements, which flags values exceeding a certain threshold and checking the validity of SM variations in relation to changes in soil temperature and precipitation (Dorigo et al., 2013). A second flagging procedure analyses the shape of the SM time series to detect outliers, positive and negative breaks, saturation of the signal, and unresponsive sensors. In this analysis, only values with the flag "G" (good) are kept. Dorigo et al. (2021) reported on the recent developments within the ISMN and provide a detailed documentation on the network update activities and data quality curation. This includes an ongoing activity on quality curation of historic time series that aims for most consistent and reliable in situ data. Consequently, the available time series data at ISMN may be altered over time in terms of flagging and value density. To ensure a long-term funding, the ISMN operations were transferred by the end of 2022 to the

International Center for Water Resources and Global Change (ICWRGC) and the German Federal Institute of Hydrology (BfG), and both new hosting intuitions are dedicated to continue ongoing activities on network integration and quality check.

Spatial representativeness is an issue when comparing remotely sensed and in-situ data. This metric indicates how well an in-situ observation, which is essentially a point measurement, represents the data gathered by the much larger satellite footprint. This is also different for each station and depends on topography, soil types and microclimates. The geographic distribution of the available stations in the current validation period is shown in Figure 2, illustrating the inhomogeneous global distribution of ground data, leaving many parts of the world unobserved.

For more information on the ISMN network details and contacts, please visit: https://ismn.earth/en/networks/.

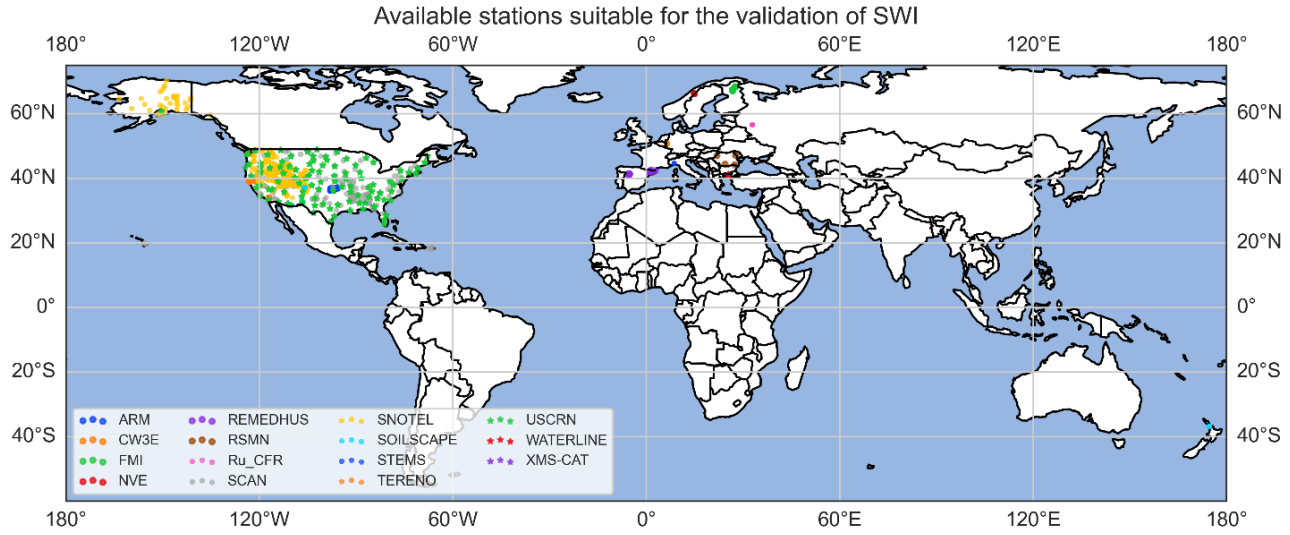


Figure 2: Map of used ISMN networks, showing all stations with data available in the period 2007-2024. Only networks with data points in 2024 are displayed.

The ISMN networks are constantly updated, either via an Application Programming Interface (API) automatically refreshing the data in near-real-time, or via manual updates. Some networks have significant update delays and were not updated for entire 2024 at the time of the validation. The ISMN data for the global analysis was accessed the 15th of April 2025.

The selected ISMN networks have different data availability and references (see Table 4), but many of the networks and stations used in VR 2023 are used in VR 2024, newly available stations are integrated, and some stations do not provide data anymore in 2024 (for this VR we have totalling *1378* stations; which is an increase compared to the VR 2023). Overall, there is again some fluctuation

in the set of used networks: the ISMN network FR_Aqui is not used in this year's VR as there is no data update for 2024, the networks CW3E, NVE, and WATERLINE are added to the analysis.

To get the listed start- and end-date, the maximum and minimum date found in any station of the network is given. It can be seen that most in-situ networks do not cover the full validation period. Furthermore, the maximum date might not apply since this could be either for a time series that was not soil moisture, or not in the right observation depth.

Additionally, some networks, namely SCAN and SNOTEL, have changed how they reported the time series for each station since our last access for VR 2024. The sensors in these networks have been replaced over time and each replacement is now reported as separate timeseries, whereas in previous years they were merged temporally to a single timeseries. This not only led to significant increase in number of individual timeseries related to these stations, but it also increased the number of timeseries per station, as non-overlapping timeseries related to the same sensor was reported separately. To mitigate this change and harmonize the data with the rest of the networks, as well as previous years reports, each split timeseries are merged back together and considered as one station.

Table 4: Data availability and references for the used ISMN in-situ networks.

Network Name	*Nr of stations*	Start	End	Reference
ARM	*204*	2007-01-01	2025-01-01	http://www.arm.gov/
CW3E	*36*	2017-06-29	2025-01-01	https://cw3e.ucsd.edu/
FMI	*27*	2007-01-25	2025-01-01	http://fmiarc.fmi.fi/
NVE	*6*	2012-01-01	2025-01-01	https://www.nve.no/hydrology/
RHEMEDUS	*24*	2007-01-01	2025-01-01	http://campus.usal.es/~hidrus/
RSMN	*20*	2014-04-09	2025-01-01	http://assimo.meteoromania.ro/
Ru-CFR	*6*	2015-05-25	2025-01-01	https://ismn.earth/en/networks/?id=Ru_CFR
SCAN	*208*	2007-01-01	2025-01-01	http://www.wcc.nrcs.usda.gov/scan/
SOILSCAPE	*188*	2011-08-03	2024-12-17	https://soilscape.usc.edu/
SNOTEL	*419*	2007-01-01	2024-01-01	http://www.wcc.nrcs.usda.gov/, Leavesley et al., 2008
STEMS	*4*	2015-12-04	2025-01-01	https://sustag.to.cnr.it/index.php/cannona-db
TERENO	*14*	2009-12-31	2025-01-01	https://www.tereno.net/joomla4/index.php
USCRN	*222*	2007-01-01	2025-01-01	http://www.ncdc.noaa.gov/crn/, Bell et al., 2013, Diamond et al., 2013

Network Name	*Nr of stations*	Start	End	Reference
WATERLINE	*3*	2021-06-10	2024-09-09	https://www.waterlineproject.eu/
XMS-CAT	*21*	2016-08-01	2025-01-01	https://visors.icgc.cat/mesurasols/#9/42.1765/1.1132
sum of stations	*1378*			

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4.5 OTHER REFERENCE PRODUCTS

For the regional analysis, the anomalies from multiple products are compared to SWI anomalies (as described in Section 4.2).

For the drought event in Central Europe, the Fraction of Absorbed Photosynthetically Active Radiation (fAPAR) product is utilized and obtained from the Global Drought Observatory of Copernicus[3]. The product is derived from the Visible Infrared Imaging Radiometer Suite and indicates the fraction of solar radiation captured by photosynthesis and the reasoning behind is that precipitation deficit can result in changes in absorbed photosynthetic radiation by plants. The fAPAR anomaly dataset comes as 10-days interval raster files. The period comprised for the anomaly calculation ranges from 2012 until the last available full year (2024 for our study). See also (Global Drought Observatory - Copernicus, 2023).

Moreover, all three regional analyses use the Standardized Precipitation Index (SPI), which measures precipitation anomalies by comparing total precipitation for a defined length of time (SPI-12 months is used in this report) against long-term records of rainfall (1981 to 2010). A probability gamma distribution is fitted on the 30 years of precipitation data. The cumulative probability is derived and standardised from the monthly cumulative rainfall observations. Input precipitations are retrieved from daily in-situ precipitation measurements aggregated into monthly totals over Europe (Surface Synoptic Observations (SYNOP), operated by JRC, (Toreti, 2014), and monthly gridded precipitation from the Global Precipitation Climatology Center (Schneider et al, 2015).

In the flood events in South America, the following products from the study of Simoes-Souza et al. (2025) were used: 10-meter air temperature anomaly, precipitation water vapour anomaly, anomalous height at 500-hPa and winds, and mean precipitation. The first three metrics are obtained from the Modern Era Retrospective Analysis for Research and Applications version 2 (MERRA-2) product produced by NASA's Global Modeling and Assimilation Office (GMAO), which provides data from 1980 to present (Gelaro, 2017). Simoes-Souza et al. (2025) computed the percentiles for each grid point for April and May across the entire 40-year time range of MERRA-2 to produce the final images. The last metric (mean precipitation) is obtained by the IMERG data, which stands for Integrated Multi-satellitE Retrievals for Global Precipitation Measurement. It is an algorithm that combines information from the Global Precipitation Measurement (GPM) satellite constellation to estimate precipitation over the majority of the Earth's surface and provides information since year

[3] https://drought.emergency.copernicus.eu/tumbo/gdo/download/ (accessed in May 2025)

2000 to the present (National Integrated Drought Information System, Accessed on June 2025). For the study in Simoes-Souza et al. (2025), 24 years were used for the calculations.

Anomalous heigh at 500-hPa product refers to atmospheric circulation anomalies (how much it differs from the average values) that are characterized using the 500-hPa geopotential height. This variable, commonly used in meteorology, can be defined as the distance up in the atmosphere before the air pressure drops to 500 hPa, which is usually around 5 to 5.5 km above sea level (Atmospheric Circulation - Copernicus, 2022). With the intention to detect atmospheric blockings, the data producer used the metric by scaling the anomalous geopotential height at 500 hPa with a latitude-dependent factor ($sin\ (45°)\ /sin\ (latitude)$), and identifying periods where the adjusted value exceeds 100 m for more than 10 consecutive days. Moreover, a 5-day running-averaged time series was applied to reduce the influence of short-term variability (Simoes-Sousa et al., 2025).

Finally, for the eastern Africa use case, the monthly rainfall anomalies from the Kenya Meteorological department (KMD) yearly assessment (Nying'uro, 2024) supported the evaluation.

4.6 DATA AVAILABILITY

Table 5 shows an overview on the data availability of the SWI and the reference datasets. The GLDAS Noah model data covers the complete validation period, whereas some of the ISMN networks do not cover the full validation period.

Table 5: Available datasets for validation

Dataset	Start	End	Spatial Coverage, Resolution	Observation depth	Unit
SWI V3 NRT	2024-01-01	2024-12-31	Global, 25km	T-value	Percent
SWI V3 archive	2007-01-01	2023-12-31	Global, 25km	T-value	Percent
GLDAS	2007-01-01	2024-12-31	Global, 0.25°	0.0 – 0.1 m 0.4 – 1.0 m	°C and Kg/m²
ISMN	different, min 2007-01-01	different, max 2024-12-31	*1378* stations	0.0 – 0.1 m 0.5 m	m^3/m^3

4.7 SOIL LAYER COMPARISON MATRIX

The different T-values of the SWI products were compared to different layers of the GLDAS model as well as in-situ data. This was done since higher T-values represent deeper soil layers. Table 6 shows which T-values were attributed in the comparison to which layer depths of the reference data. General recommendations and insights on the attribution of T-values to soil depth are given in the SWI PUM document [CGLOPS1_PUM_SWIV3-SWI10-SWI-TS].

Table 6: Which T-values were compared to which layer of the modelled and in-situ datasets

SWI T-Value	GLDAS layer depth	In-situ observation depth
1, 5, 10, 15, 20, 40	0.0 – 0.1 m	0.0 – 0.1 m
60, 100	0.4 – 1.0 m	0.5 m

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5 RESULTS

5.1 GLOBAL ANALYSIS

The comparisons were done for all eight T-Values of the CLMS SWIv3 product, but we will show in this report only the results for T=1, 20 and 100 in detail, since nothing out of the ordinary was detected for the other T-Values.

The results are ordered first by reference dataset and second by metric, and are presented as images as well as in tabular form. The tables show the mean and median values of the computed metrics for the validation period, as well as 50th-, 70th- and 80th- percentiles of the absolute differences between the current validation period and the reference period. For example, the 70th percentile means for example that 70% of all differences are smaller than the given value, in absolute (positive) domain. All images map globally the metric measured during the reference period as well as during the current validation period. Additionally, the difference (reference – current) is mapped to illustrate the development in the current year.

The coloured histograms below the maps show the distribution of the data values (omitting statistically non-significant results that lead to the gap around zero correlation), as well as the median and the lower- and upper- quartiles as vertical lines.

To avoid statistical inconsistencies from common global projections (featuring typically oversampling in high latitudes), the data is processed on a Discrete Global Grid (DGG) with a regular point spacing of 12.5 km. This is following the gridding approach used for the SWI-TS product (see [GIOGL1_ATBD_SWIV3]).

5.1.1 Comparison with GLDAS Noah

5.1.1.1 Results for SWI

5.1.1.1.1 Pearson's correlation coefficient rho

The results for Pearson's correlation coefficient rho (Figure 3 to Figure 5) show that correlations are different at the 95% confidence interval for some areas. Statistical significance means that the calculated confidence intervals do not overlap.

Most importantly, the results for the reference period and 2024 show very similar patterns, indicating a stable performance of the SWI product, not suffering from obvious substantial or systematic quality degradation. In the histograms, one can see that rho-values for the current validation period are slightly more extreme, as that the distribution in the histograms is more stretched. Areas with positive rho-values tend to be more positive and areas with negative rho-values tend to be more negative. This

behaviour was already observed in the previous years and is likely due to the shorter observation period of only one year.

Again in 2024, over desert areas, e. g. over the Sahara, Atacama, the Namibian deserts, or the Arabian Peninsula, and also over the semi-arid areas in Iran, USA, and Mexico the SWI shows weak or no agreement with the GLDAS model. This confirms findings in the literature that the ASCAT data does not model soil moisture correctly in very arid areas, where the model's basic assumption on a linear relationship between backscatter and soil moisture does not apply anymore (McColl et al., 2014, Wagner et al., 2022).

However, focusing now on areas with generally good agreement, large parts of the Earth show a slightly improved agreement with the model in 2024 than in the reference period. Large parts of North America, Siberia, and North-eastern Europe, and parts of China show better agreement in 2024 than the previous years (blue in Figure 3, lower right), while Mid-western USA, Mexico, Western Europe, Southern Africa, Northern Africa, and sections of Australia show worse performance.

The decrease over Western Europe resembles patterns of areas with presumably wrong wetting trends in the ASCAT data, as has been reported recently by users. To what extent this is due to effects from urban growth and Radio Frequency Interference (RFI) on the parametrisation of the input ASCAT SSM data, is to be investigated in the future. Particularly the RFI issue is a very recent development, and is subject to present scientific discussion.

When looking on the deeper soil layers (T=100), decrease in agreement dominates the difference map, particularly over North America, central Australia, and large parts of Asia. The distinct growth of poor agreement along the Taiga sections of North America and Eurasia should be closely monitored in upcoming analyses.

Overall, the spatial patterns between the reference period and 2024 are well-known, with year-to-year deviations similar to what was obtained in previous VR analyses, so a consistent quality deprecation of the SWIv3 product in 2024 is unlikely from this data. Looking on the histograms for the current and reference periods in Figure 3 to Figure 5, the median rho values are consistently higher for the current period (2024) for two T-values (T=1, T=20), whereas only for T=100 the median and lower quartile went down. The latter observation is congruent with the widespread occurrence of strong negative correlation values mapped in Figure 5 (top).

Looking on the histograms collecting the significant differences between correlations of reference and current period, all three (T=1/20/100) lean towards the negative side, indicating that the majority of pixels show a small or no significant increase - but as the median values are in the positive sections - more than 50% of the pixels feature significant lower agreement than in the reference period, with

some of them show a large decrease (larger than diff_rho=0.5). Again, this is particularly the case for T=100, with many pixels agreeing less with GLDAS in 2024.

The median values as well as the differences shown in Table 7 compare well with the insights gained from the previous exercise performed on 2023 data, but with lower mean correlations, with decrease for the mean of T=20 from 0.36 to 0.34, and a decrease for T=100 from 0.25 to 0.20 The median values of 0.55, 0.50, and 0.37 in Table 7 correspond to the dashed lines in the histograms for the current period in Figure 3 to Figure 5. To put the changes in perspective, it is important to remember that although a correlation coefficient quantifies the strength of a linear relationship, its stability and interpretability are strongly dependent on the sample size. Smaller samples (like in this case from one single year) are more prone to variability. Larger sample sizes like 17 years in this case, by contrast, enable a more precise estimation and stronger inferential power.

The 50th/70th/80th percentiles of the absolute values of the significant differences are on average the same for T=1, and lower for T=20 and T=100 than in 2023, meaning that there are overall smaller absolute differences between the current and reference period.

As statistics for 2024 are within the small and regular fluctuation experienced in the previous annual VR reports, a continuation of a stable performance of the SWIv3 product can be assessed. The widespread occurrence of less well agreeing T=100 time series in the cold climates should be monitored in upcoming analyses, when the upgraded cold masking is effective.

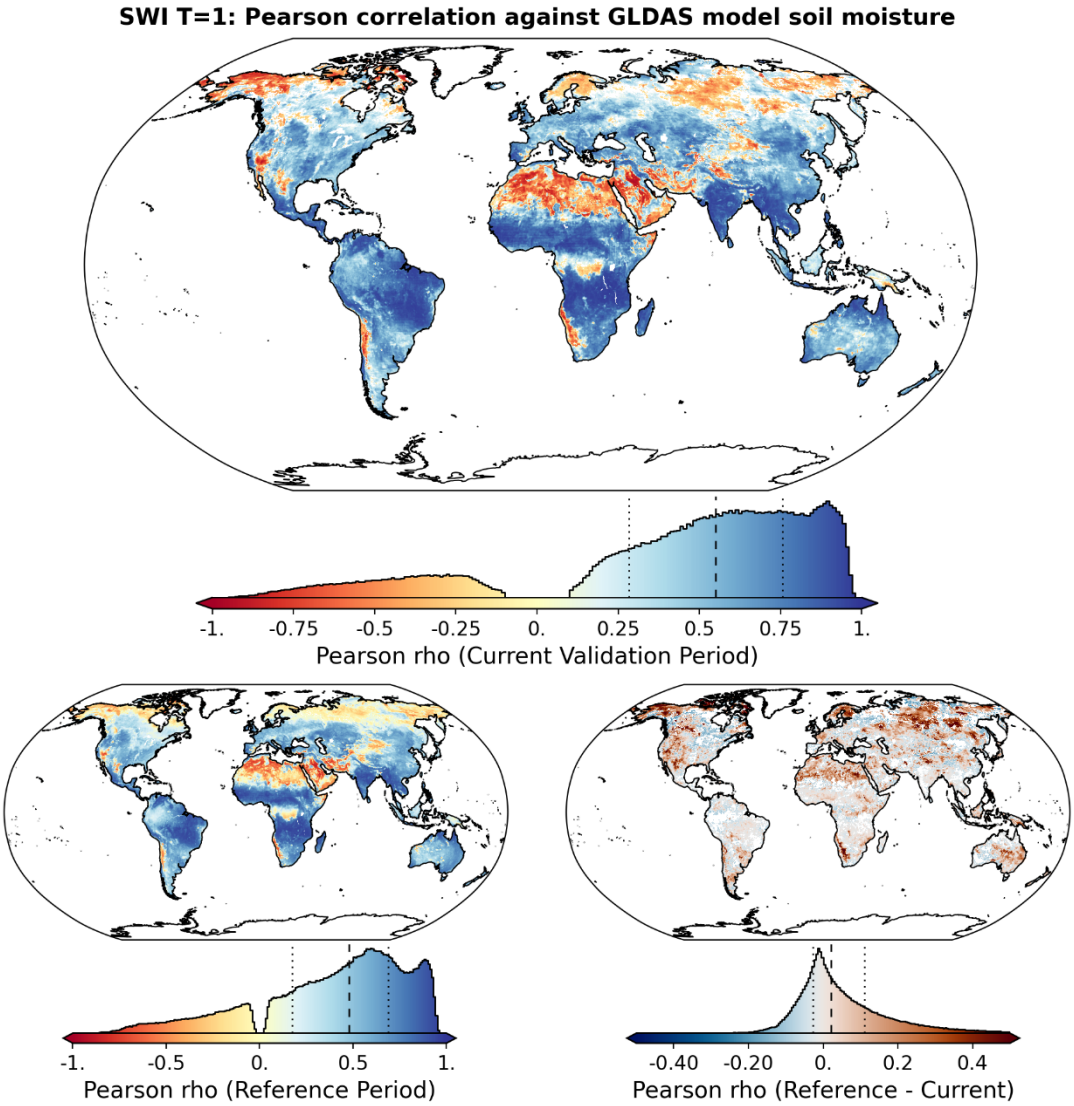


Figure 3: Pearson's correlation coefficient rho between SWI T=1 and GLDAS Noah, as well the difference between the reference period (2007-01-01 until 2023-12-31) and the current validation period (2024-01-01 until 2024-12-31). Only the statistically significant differences are plotted. The histograms plots show the data from the maps above as distribution, as well as the median (dashed line) and the lower and upper quartile (dotted lines).

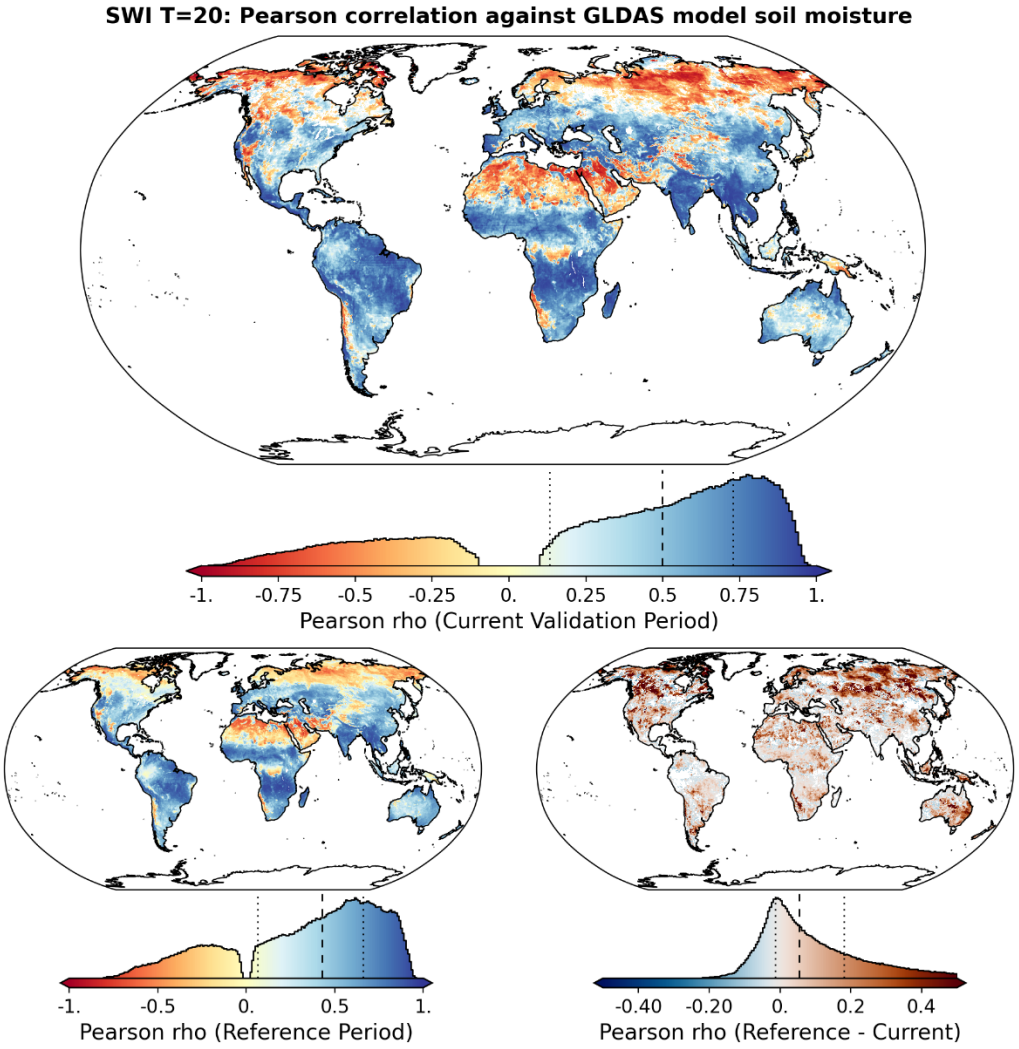


Figure 4: Pearson's correlation coefficient rho between SWI T=20 and GLDAS Noah, as well the difference between the reference period (2007-01-01 until 2023-12-31) and the current validation period (2024-01-01 until 2024-12-31). Only the statistically significant differences are plotted. The histograms plots show the data from the maps above as distribution, as well as the median (dashed line) and the lower and upper quartile (dotted lines).

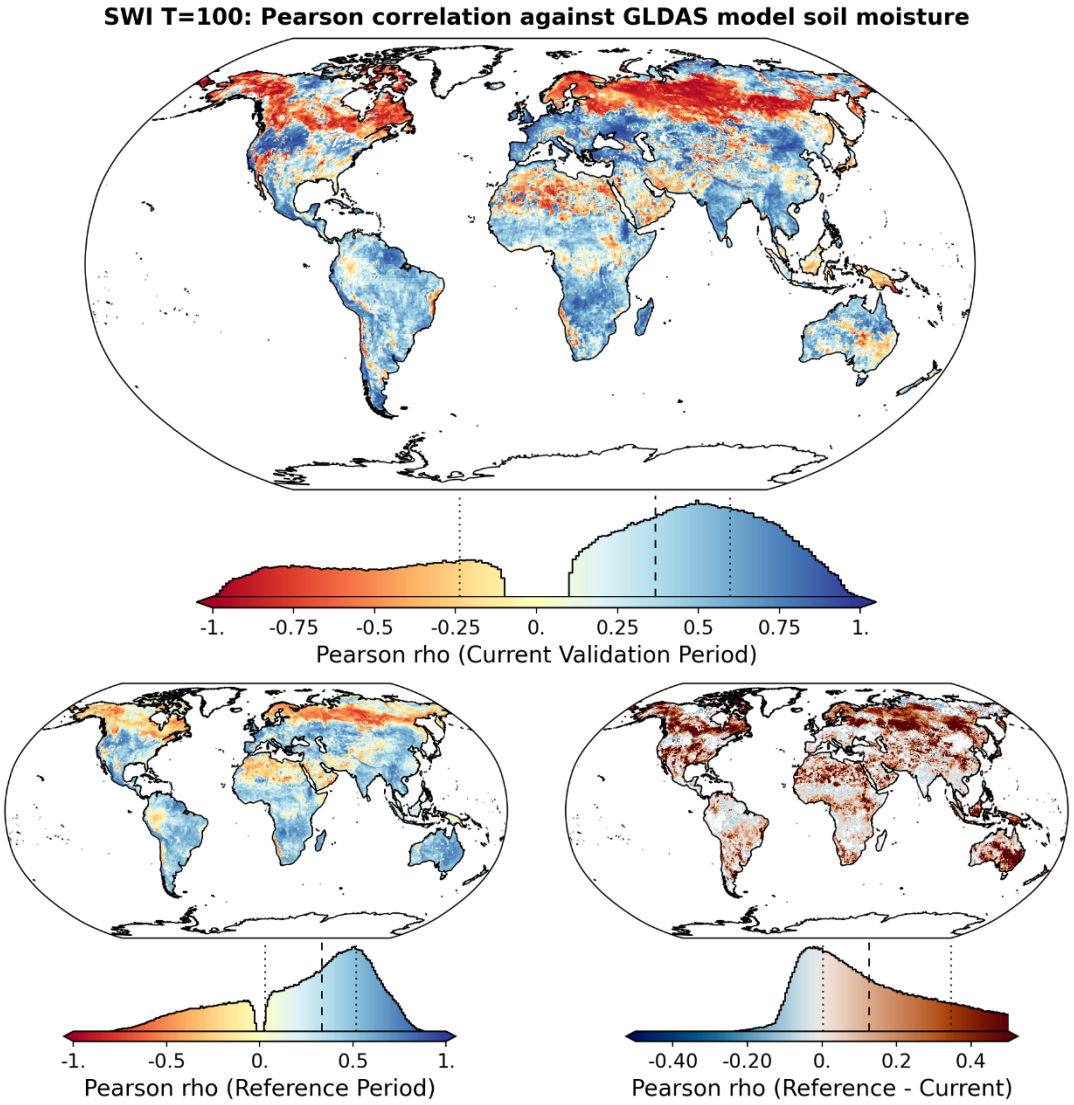


Figure 5: Pearson's correlation coefficient rho between SWI T=100 and GLDAS Noah, as well the difference between the reference period (2007-01-01 until 2023-12-31) and the current validation period (2024-01-01 until 2024-12-31). Only the statistically significant differences are plotted. The histograms plots show the data from the maps above as distribution, as well as the median (dashed line) and the lower and upper quartile (dotted lines).

Table 7: Mean / median rho between SWI and GLDAS Noah, as well as percentiles of the absolute values of the significant differences. All values in m^3/m^3.

	rho in current period		*absolute values of difference in rho*		
	mean	**median**	**50th percentile**	**70th percentile**	**80th percentile**
SWI T=1	0.43	0.55	0.15	0.21	0.26
SWI T=20	0.34	0.50	0.18	0.25	0.31
SWI T=100	0.20	0.37	0.25	0.35	0.44

Figure 6 shows the quarterly rho-values over selected regions as shown in Figure 1. The yearly evolution of the performance over eastern Africa, and Contiguous USA including the separately analysed California, is similar for the reference and the current period, showing overall comparable annual SWI dynamics in these regions.

The plots show for yearly quarters Q1-Q4 the (seasonal) Pearson's correlation coefficient rho of agreement between the SWIv3 and the GLDAS model. Most notable deviations between the 2024 and 2007-2023 performances are improvements over Italy and Patagonia in Q3-Q4. All observed (current) 2024's deviations of the six regions are within the 1 standard deviation range from the 2007-2023 reference and show an overall positive trend.

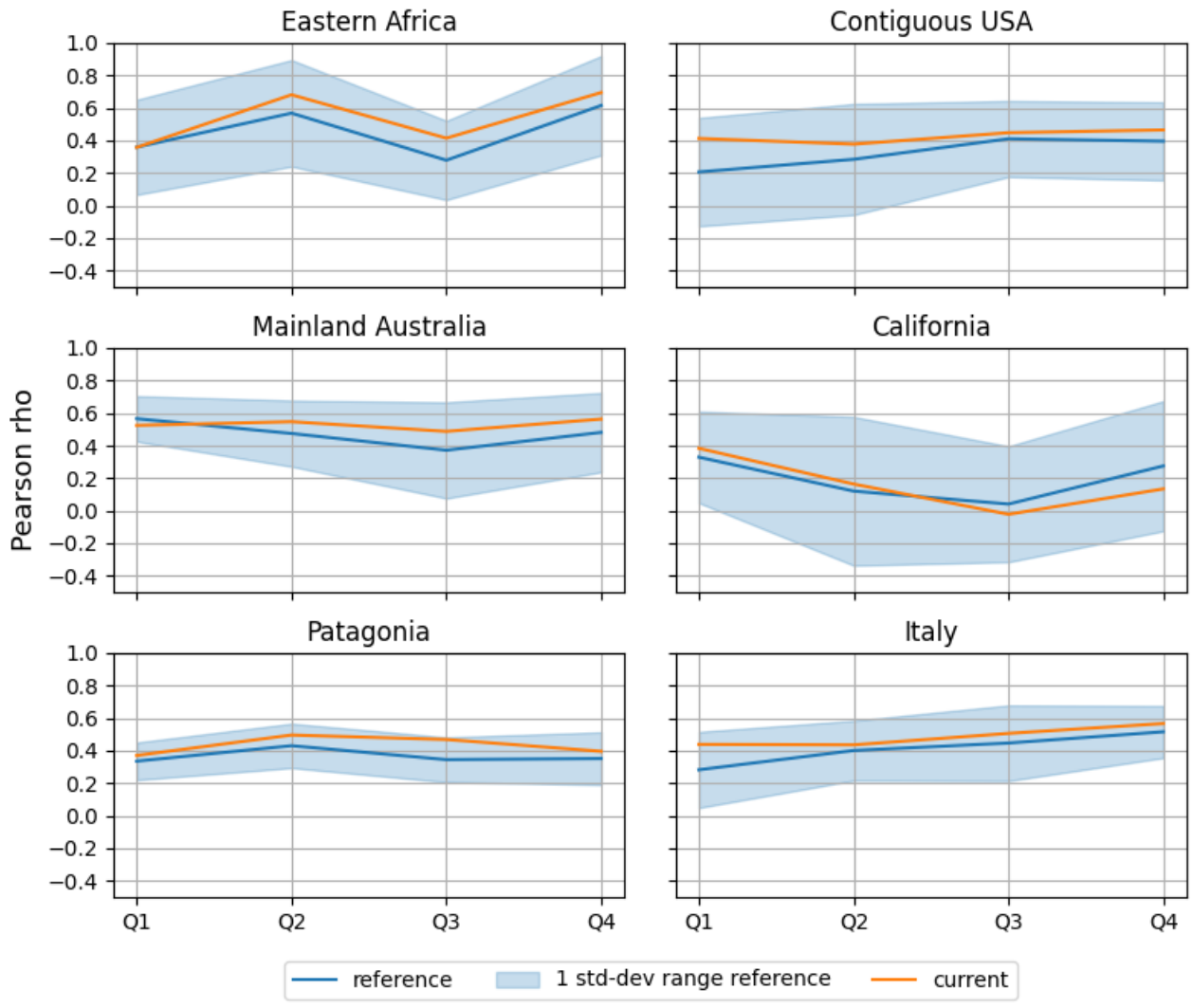


Figure 6: Average 3-month Pearson's correlation coefficient rho between SWI T=1 and GLDAS Noah, for the reference period (2007-01-01 until 2023-12-31, blue line) and the current validation period (2024-01-01 until 2024-12-31, orange line) over various areas. The standard deviation for the reference is shown as light blue surface.

5.1.1.1.2 Root mean square difference

Figure 7 to Figure 9 show the RMSD-results for the two validation periods and their differences on maps with associated histograms. Again, there seem to be no strong systemic, global-scale differences. However, over many world regions the RMSD has improved to lower values, e. g. over northern Canada, central Africa, South America, and large parts of Asia. When consulting previous exercises results, most differences can be interpreted as year-to-year fluctuations. Higher RMSDs are observed this year particularly over northern Australia, Mid-western USA, eastern South America, Central Africa, the Arabian Peninsula, and South Asia for T=1, Alaska, Midwest USA, eastern South America, central and north-western Africa, the Arabian Peninsula, India, Eastern Europe and Russia, and northern Australia for T=20 as well as eastern USA and Canada, eastern South America, Eastern Europe and Russian, south-eastern Africa, India, and northern Australia particularly for T=100.

For T=1, median RMSD is lower in 2024 than 2007-2023, and the distribution of the significant difference leans to positive, with smaller mean RMSD and more than 50% smaller RMSD than in the reference period, continuing the trend from SQE/VRs 2017-23.

For T=20, median RMSD is slightly higher in 2024 than 2007-2023, and in the distribution of the significant differences, slightly more than 50% of the pixels show higher RMSD.

For T=100, very marked difference map is plotted, also reflected in the larger share of strong RMSD increases (red values in Figure 9 bottom right).

Table 8 shows the mean and median values of the RMSD for the validation period (year 2024), as well as percentiles of the absolute values of the significant differences between the validation and reference periods. Mean values for all depths show a decrease, while the median values remain stable compared to last year's VR. Like last year, the percentiles of the absolute differences between the validation and reference periods are around 0.01 m^3/m^3, suggesting an overall stable performance of the SWI data.

Table 8: Mean / median RMSD between SWI and GLDAS Noah, as well as percentiles of the absolute values of the differences. All values in m^3/m^3.

	RMSD in current period		*absolute values of difference in RMSD*		
	mean	**median**	**50th percentile**	**70th percentile**	**80th percentile**
SWI T=1	0.05	0.05	0.01	0.01	0.01
SWI T=20	0.05	0.05	0.01	0.01	0.02
SWI T=100	0.05	0.05	0.01	0.01	0.02

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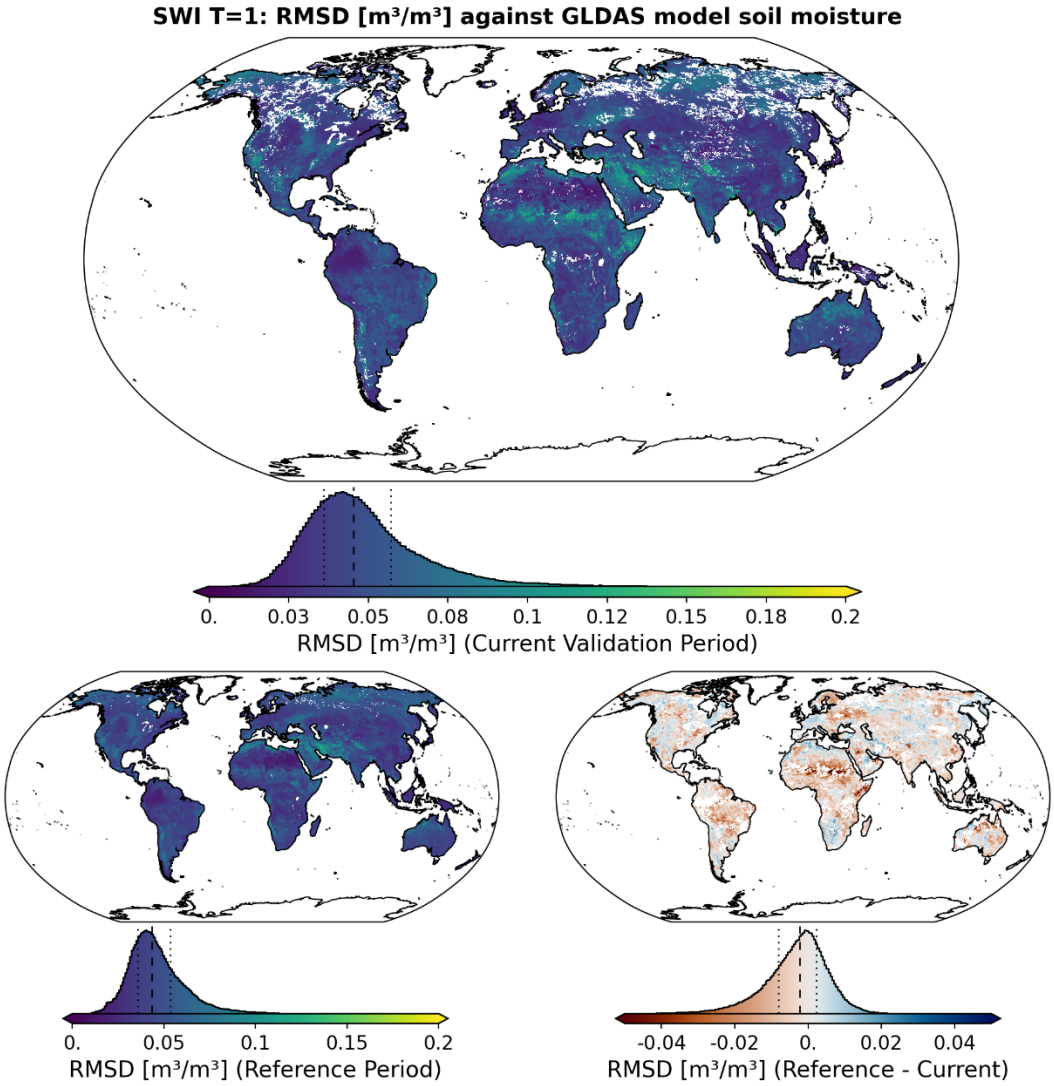


Figure 7: RMSD between SWI T=1 and GLDAS Noah, as well the difference between the reference period (2007-01-01 until 2023-12-31) and the current validation period (2024-01-01 until 2024-12-31). Only the statistically significant differences are plotted. The histograms plots show the data from the maps above as distribution, as well as the median (dashed line) and the lower and upper quartile (dotted lines).

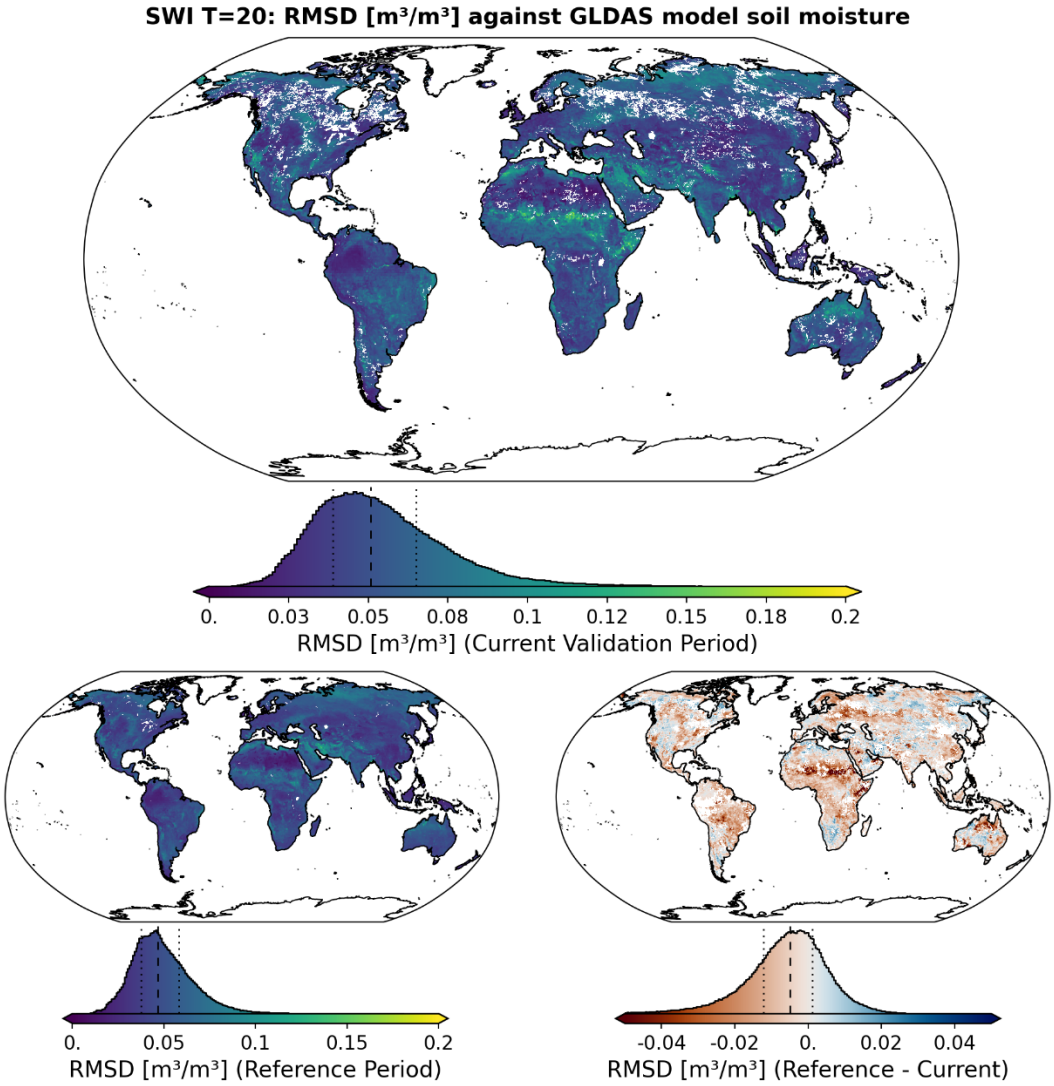


Figure 8: RMSD between SWI T=20 and GLDAS Noah, as well the difference between the reference period (2007-01-01 until 2023-12-31) and the current validation period (2024-01-01 until 2024-12-31). Only the statistically significant differences are plotted. The histograms plots show the data from the maps above as distribution, as well as the median (dashed line) and the lower and upper quartile (dotted lines).

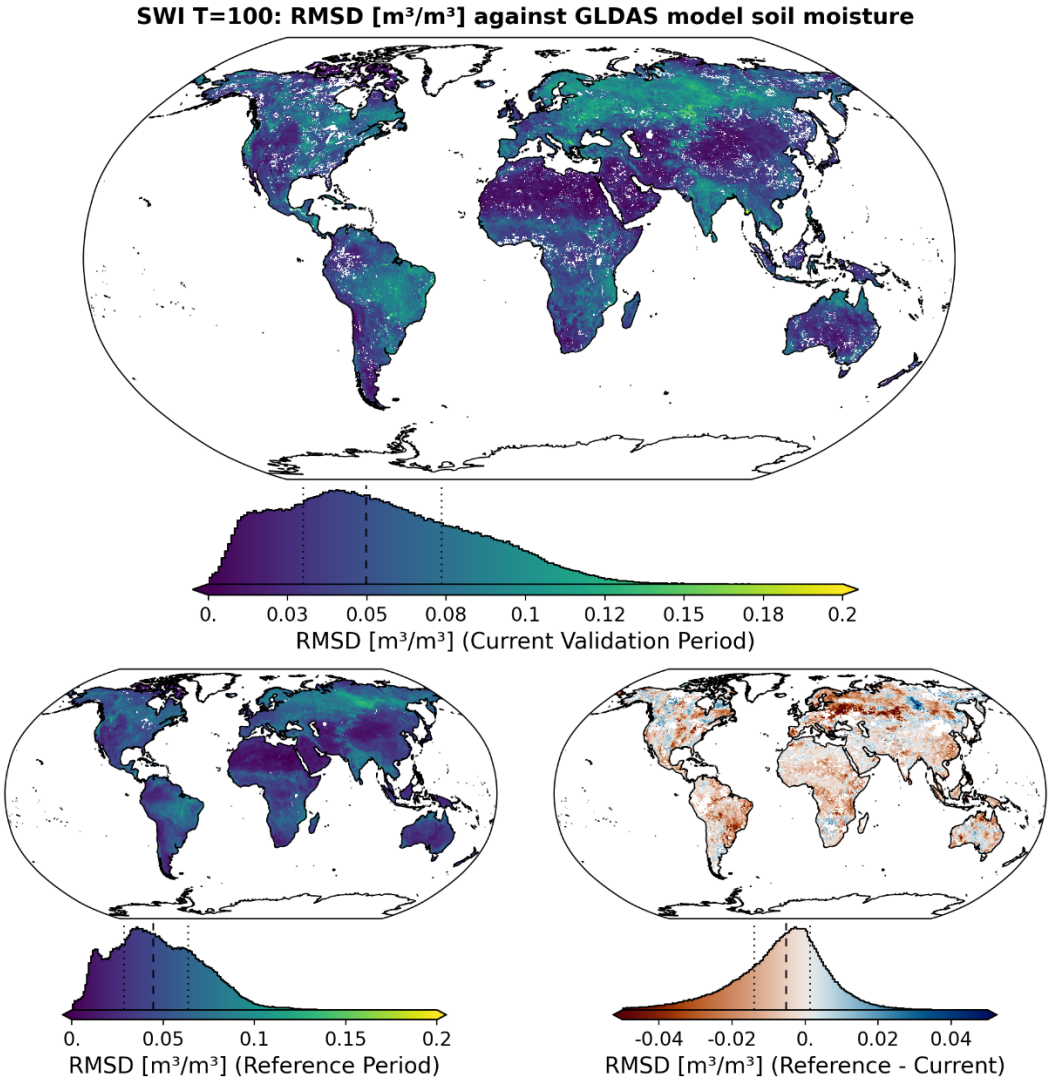


Figure 9: RMSD between SWI T=100 and GLDAS Noah, as well the difference between the reference period (2007-01-01 until 2023-12-31) and the current validation period (2024-01-01 until 2024-12-31). Only the statistically significant differences are plotted. The histograms plots show the data from the maps above as distribution, as well as the median (dashed line) and the lower and upper quartile (dotted lines).

5.1.1.2 Results for Surface State Flag

The SSF was evaluated against the soil temperature data from GLDAS. Figure 10, Figure 11, and Figure 12 show the percentage of correct SSF classifications, and the maximum and mean in consecutive wrongly flagged days per grid point, respectively.

The spatial patterns for the percentage of correct SSF are globally almost identical for the current and reference period, as well as with both results from SQE/VRs 2018-2023. It is recognised that the observed patterns of correct classification are relatively stable, but in 2024 a decrease is spotted over sections of Central Asia, central North America, Scandinavia, and Patagonia.

Looking on Figure 11 and Figure 12, patterns for areas showing on average many consecutive days wrongly classified remain also very stable, with longest "wrong" periods in Tibetan Plateau, Central Asia, western Europe, the Andes, but only short periods in most of the rest of the world.

We note that the obtained values of consecutive days wrongly classified are considerably lower than in the VRs before 2023. This is a result from the updated method for selecting the reference data, where now the daily minimum temperature is selected, a crucial indicator for freeze/thawing processes. Previously, the temperature values closest to the 12h UTC timestamp was selected, and this was not properly related to the local satellite overpass/measurement time. The new approach is more stable, as effects from the diurnal cycle are circumvented. This also affects the mean and median statistics in the summary Table 9.

The histogram shown at Figure 12's top forms a peculiar pattern of a high share in round integer numbers, e.g. for 1, 2, 3, etc. The histogram's values are per location the average number of consecutive days wrongly classified, and as such, can result from only a few individual periods, or from many periods of wrong classification. As many warm regions in the Subtropics undergo within the study period only few individual frost events, they contribute many periods with 1, 2, 3, etc. days length to the global statistic, whereas colder regions with more frost events yield relatively more fractional values for this metric. Hence, the histogram features peaks at low round numbers.

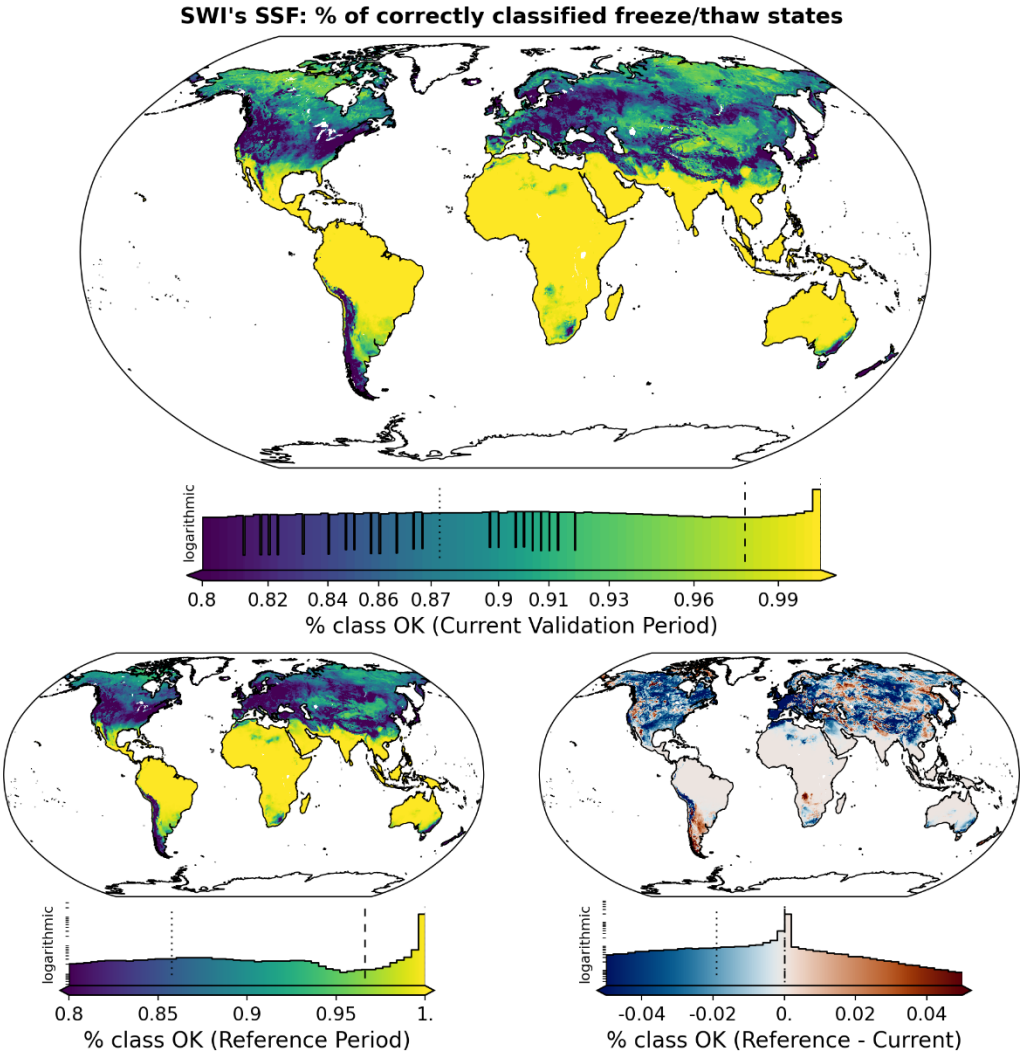


Figure 10: Percent of correctly classified freeze/thaw states compared to GLDAS Noah soil temperature as well the difference between the reference period (2007-01-01 until 2023-12-31) and the current validation period (2024-01-01 until 2024-12-31). The histograms plots show the data from the maps above as distribution, as well as the median (dashed line) and the lower and upper quartile (dotted lines). Areas without any days of SSF classification are white.

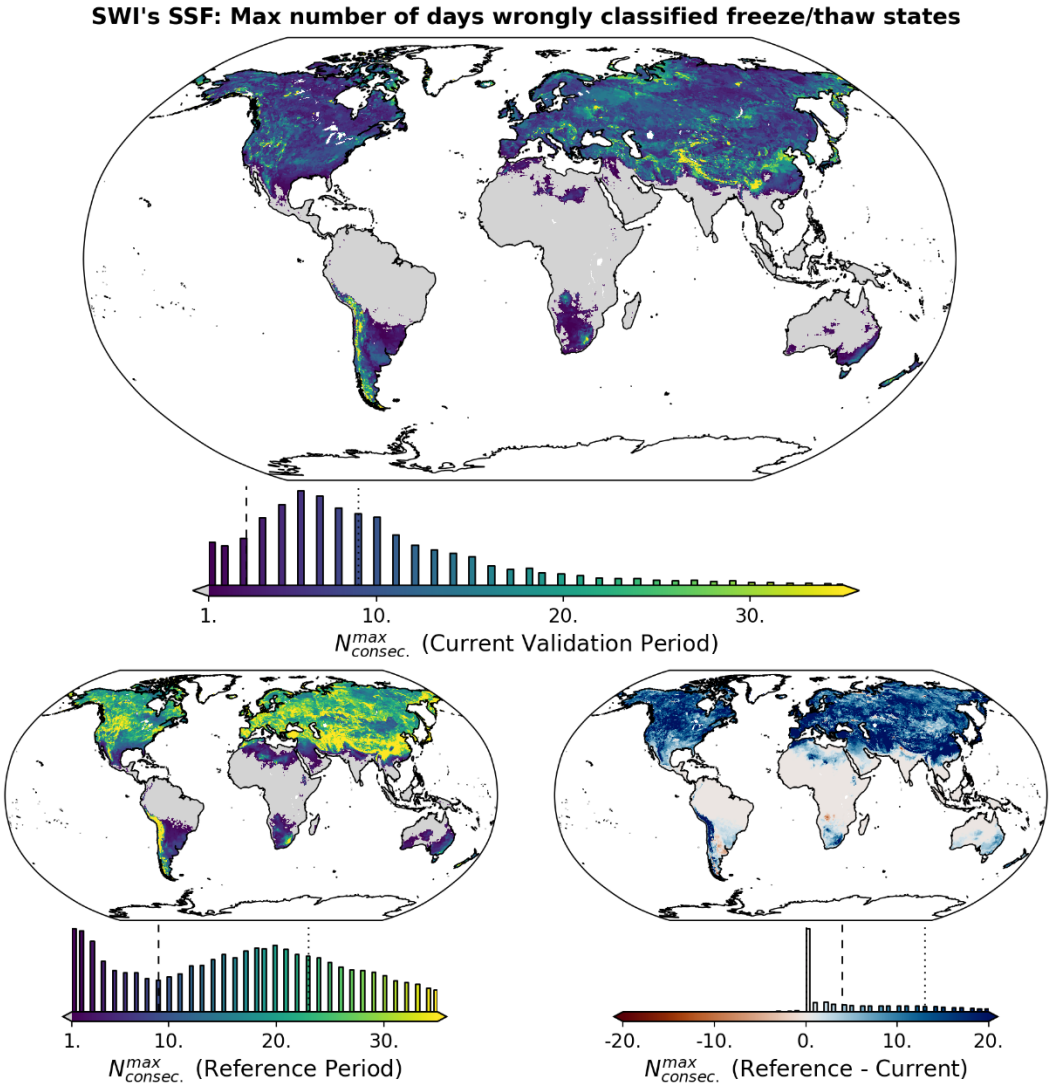


Figure 11: Maximum number of consecutive days wrongly classified when comparing the SSF against GLDAS Noah soil temperature as well the difference between the reference period (2007-01-01 until 2023-12-31) and the current validation period (2024-01-01 until 2024-12-31). The histograms plots show the data from the maps above as distribution, as well as the median (dashed line) and the lower and upper quartile (dotted lines). Areas without any days of SSF classification are white. Areas without any wrongly classified days are indicated in grey in the current- and reference plot.

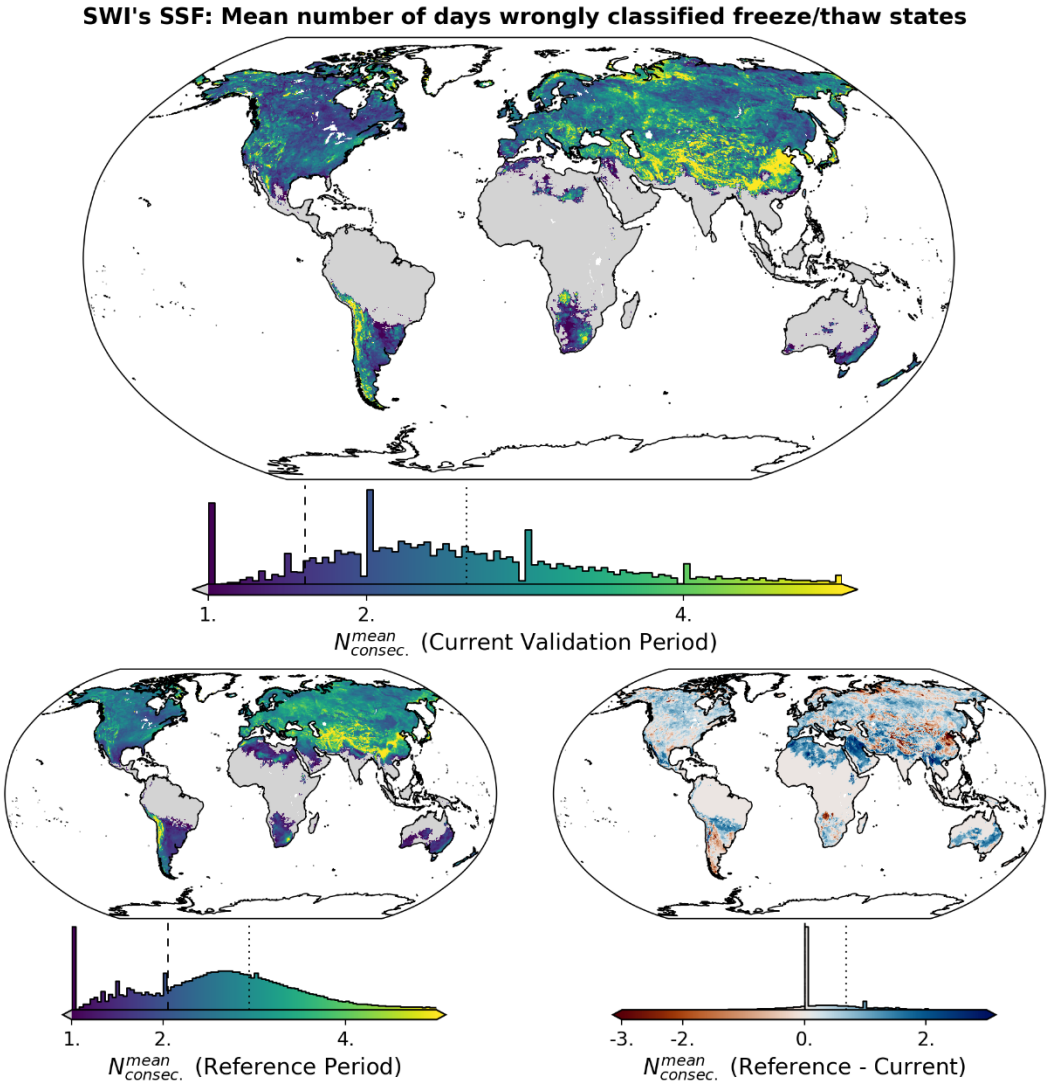


Figure 12: Mean number of consecutive days wrongly classified when comparing the SSF against GLDAS Noah soil temperature as well the difference between the reference period (2007-01-01 until 2023-12-31) and the current validation period (2024-01-01 until 2024-12-31). The histograms plots show the data from the maps above as distribution, as well as the median (dashed line) and the lower and upper quartile (dotted lines). Areas without any days of SSF classification are white. Areas without any wrongly classified days are indicated in grey in the current- and reference plot.

Table 9 summarizes the validation results for the SSF. The overall accuracy of the classification remains in 2024 very stable, as a median of 97.5% of all classifications can be considered correct. The SSF was on average wrongly set for 1.61 consecutive days with a maximum wrongly-classified period of 5.74 days.

Table 9: Mean / median metrics of the SSF validation with GLDAS Noah as well as percentiles of the absolute values of the differences.

	in current period		*absolute values of difference*		
	mean	**median**	**50th percentile**	**70th percentile**	**80th percentile**
Correct classification [%]	93.13	97.53	0.52	2.08	3.2
Incorrect classification [%]	6.87	2.47	0.52	2.08	3.2
Cons. days wrongly classified (max)	5.74	3	4	11	16
Cons. days wrongly classified (min)	0.59	1	0	0	0
Cons. days wrongly classified (median)	1.05	1	0	1	1
Cons. days wrongly classified (mean)	1.59	1.61	0.29	0.73	1.0

5.1.2 Comparison with ISMN

5.1.2.1 Results for SWI

The following plots show the results for stations with sufficient valid data in both periods (with the vast majority located in the USA), the reference years and the year 2024. Unfortunately, the plots do not show many networks in the southern hemisphere. Since the network TAHMO in Africa is not updated for 2024, the data availability for the validation is reduced in the southern hemisphere; on the other hand, the SOILSCAPE network provide now stations over New-Zealand.

5.1.2.1.1 Pearson' correlation coefficient rho

Figure 13 to Figure 15 show the rho-values and the statistically significant differences between the current validation period and the reference period.

For the global plots, to increase readability and eliminate the visual overlap of plotted stations in area with dense networks, the station-results are averaged over hexagons of 2.4° degree of diameter. This diameter was selected to visualise spatial variability of both correlation and root mean square deviation maps.

The rho-values for the current validation period are on average similar to the reference period, with better agreement in Europe, Mid-western USA and Alaska, and with worse agreement for stations in western USA.

For SWI at T-Value=1 in 2024, generally good correlations are observed over areas with stations updated for 2024. Negative or low Pearson correlations are observed in Alaska, Central-western Russia and Norway as well as West Coast USA, yet, in general the difference of correlation between the reference and validation period does not indicate significant positive or negative trends over these areas. Low correlation is expected over Norway and Alaska with challenging cold climate and the dry climate in West Coast USA is also expected to yield lower correlation as also observed in the results obtained for the GLDAS spatial correlation at the surface layer. While remaining satisfactory, slight degradation of correlation are observed over West Coast USA and Central Europe, potentially due to generally drier year over the mentioned areas. West Coast USA experienced dry winds and long wildfires for many months in 2024 and as shown in the regional study, and Central Europe experienced a significant drought in 2024. On the other hand, we observe significant correlation improvements in Spain and USA's East Coast. Spain received abnormally high precipitation in 2024, likely increasing the average soil moisture of semi-arid regions and improving the relationship between modelled and observed data. For East Coast USA, higher correlations might be attributed to the start of El Niño, leading to a wetter-than-average year for the continent, and more extreme signals caught by both datasets tend to increase correlation values.

Similar patterns are obtained over the SWI at T-Value=20 and =100, with a generally slightly lower correlation value across the in-situ stations. More spatial variability is observed over the correlation map, which can be expected when modelling deeper soil moisture with the SWI's exponential filter approach. When comparing the validation between 2024 and the validation period, a correlation decrease is observed in western North America across all three depths. The higher-than-average precipitation shows significant improvements over the Europe especially Spain in 2024 when compared to the reference period. Please, see Section 5.1.2.2 for example time series.

With the exception of the West Coast USA displaying the vast majority of stations with slightly lower correlation for the current period, the correlation values for the majority of ISMN stations are similar in 2024 (changes below 0.1 of correlation) compared to the reference period, and with many stations showing agreement with rho > 0.40, but also with more stations with (extreme) negative correlation. From the plot's distribution shown in the histogram below, it becomes clear that a small number of stations exhibit very strong negative correlations, down to rho = -0.74 (cf. with Figure 21).

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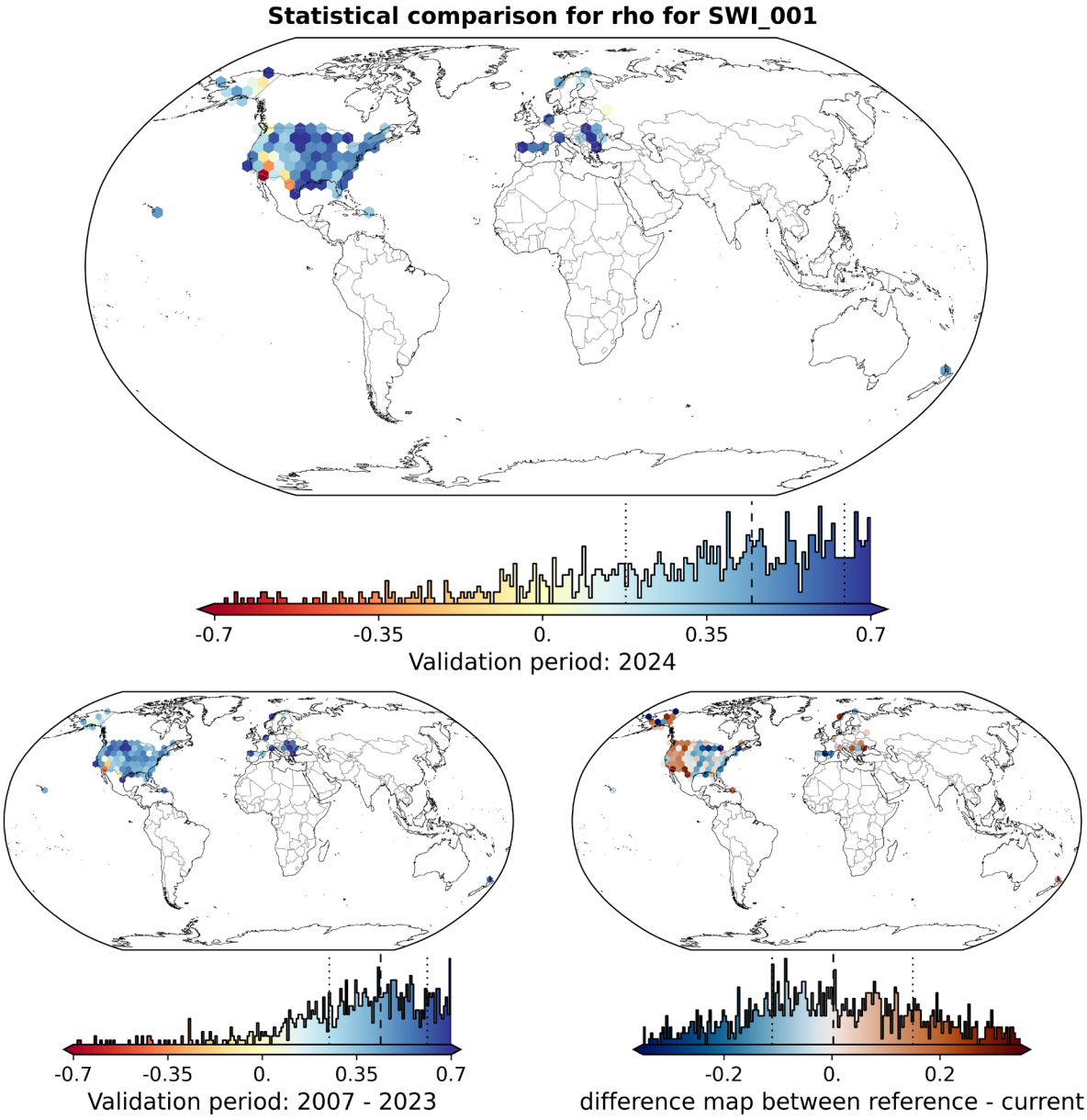


Figure 13: Pearson's correlation coefficient between SWI T=1 and in-situ data, as well the difference between the reference period (2007-01-01 until 2023-12-31) and the current validation period (2024-01-01 until 2024-12-31). In the difference map only the statistically significant differences are shown whereas the underlying distribution values show all data points. The distribution plots show the estimated kernel density plot as well as the median (dashed line) and the lower and upper quartile (dotted lines).

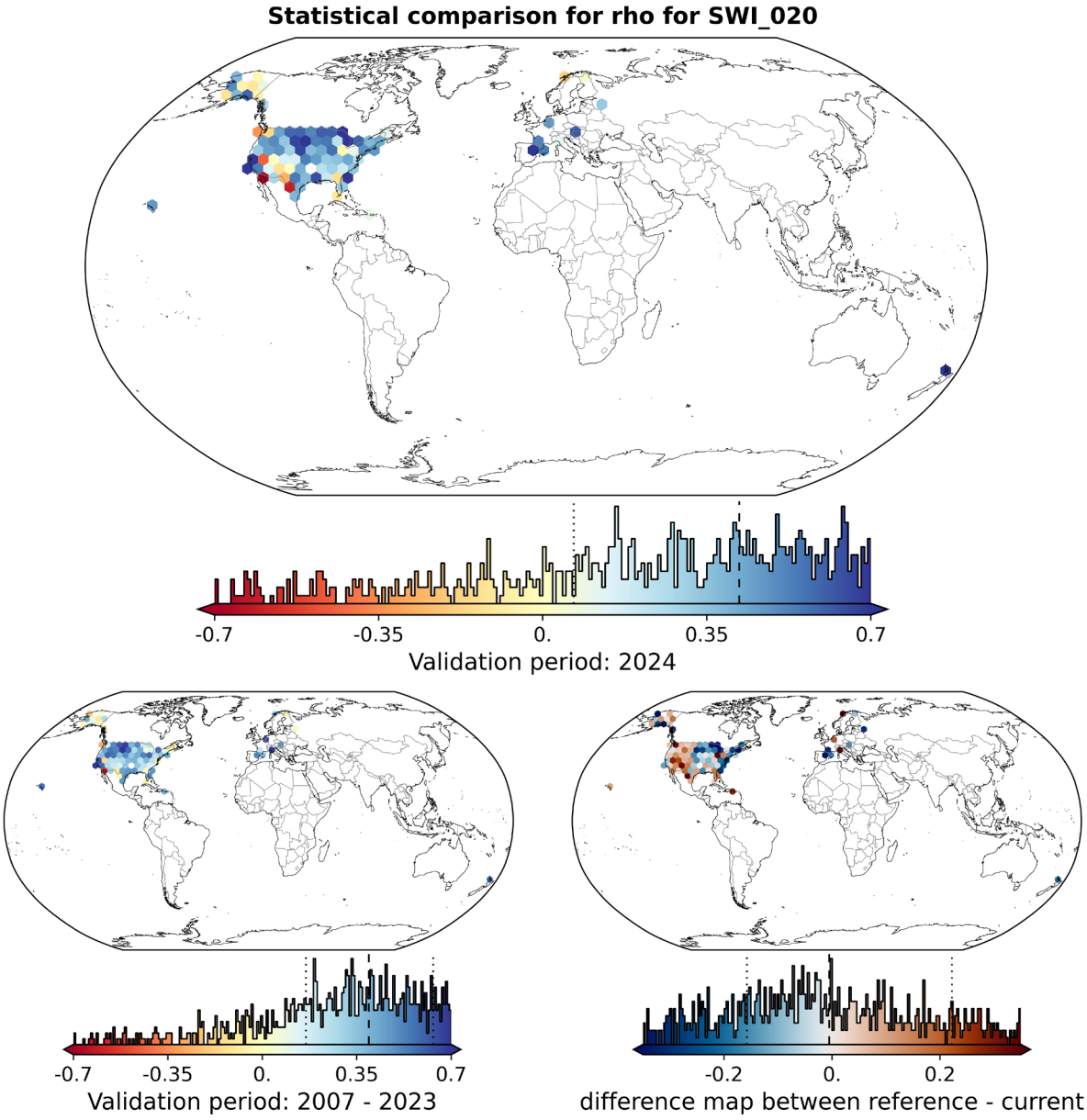


Figure 14: Pearson's correlation coefficient between SWI T=20 and in-situ data, as well the difference between the reference period (2007-01-01 until 2023-12-31) and the current validation period (2024-01-01 until 2024-12-31). In the difference map only the statistically significant differences are shown whereas the underlying distribution values show all data points. The distribution plots show the estimated kernel density plot as well as the median (dashed line) and the lower and upper quartile (dotted lines).

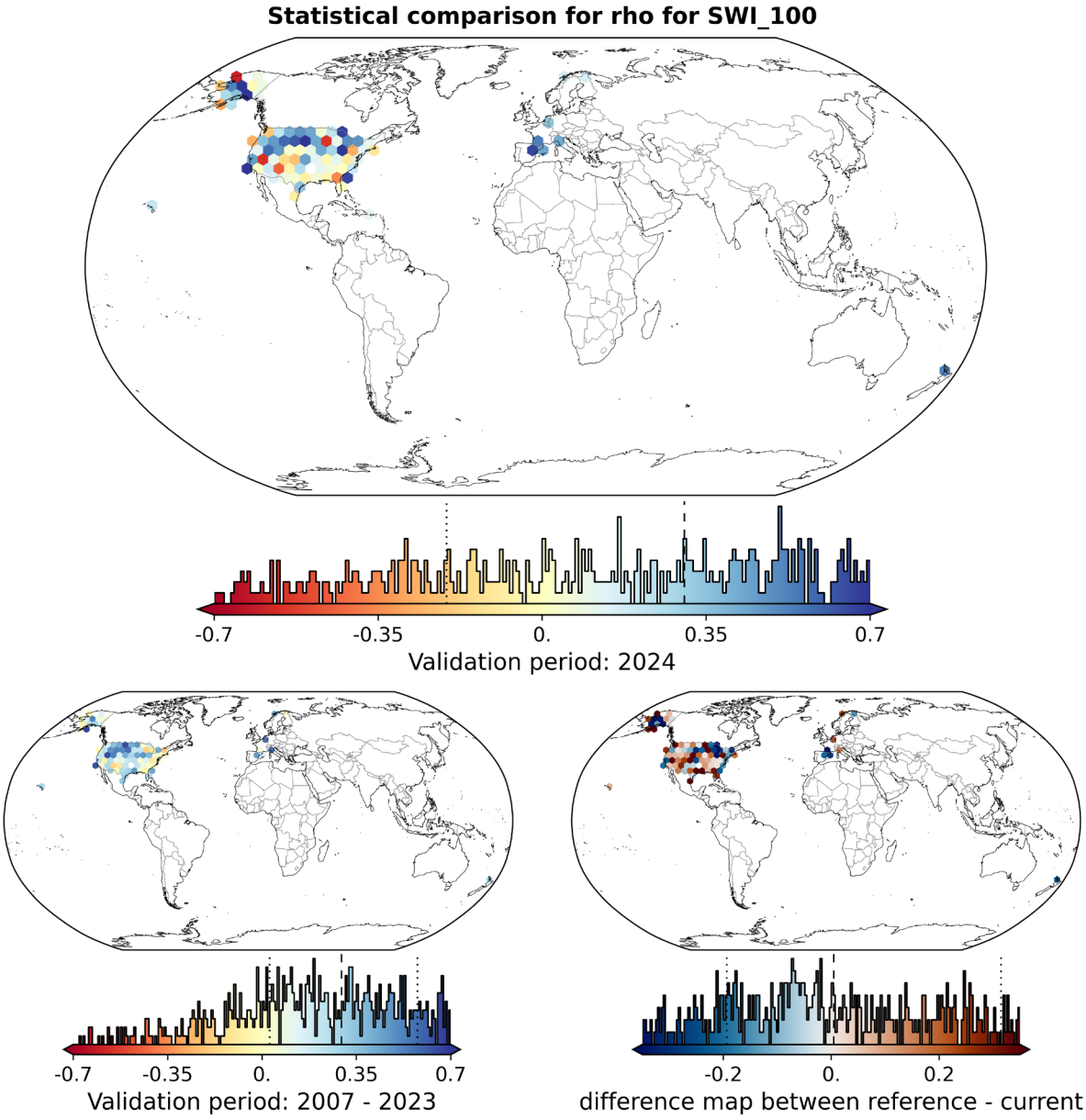


Figure 15: Pearson's correlation coefficient between SWI T=100 and in-situ data, as well the difference between the reference period (2007-01-01 until 2023-12-31) and the current validation period (2024-01-01 until 2024-12-31). In the difference map only the statistically significant differences are shown whereas the underlying distribution values show all data points. The distribution plots show the estimated kernel density plot as well as the median (dashed line) and the lower and upper quartile (dotted lines).

Summarising correlation metrics (Table 10) shows that correlation decreases with soil depth. The resulting correlations are lower than last year report, especially for T=20 and T=100. This result can

be caused by the different set of used ISMN networks, which is particularly the case for the lower soil depths where the number of available sensors is more limited. Furthermore, the temperature masking can have some effect in the overall statistics too (see Section 4.4).

Table 10: Mean / median correlation coefficient between SWI and in-situ data as well as percentiles of the absolute values of the significant differences

	rho in current period		*absolute values of differences in rho*		
	mean	**median**	**50th percentile**	**70th percentile**	**80th percentile**
SWI T=1	0.38	0.45	0.13	0.21	0.26
SWI T=20	0.33	0.42	0.18	0.30	0.39
SWI T=100	0.22	0.30	0.25	0.40	0.50

5.1.2.1.2 Root mean square difference

The RMSD maps in Figure 16 to Figure 18 all show that the RMSD are slightly higher for all depth levels in 2024 than in the multi-year reference period (the difference map has a reddish hue). Spatial patterns of the differences cannot be deduced, giving the spatially scattered distribution of negative and positive differences, however, in very broad terms, RMSD has increased for T=1 and T=20 in western USA, and for T=100 in southern USA. Looking at RMSD statistics for 2024 in Table 11, mean RMSD values have marginally increased for T=1 and T=20 and decreased for T=100 compared to last year, while the median values have stayed stable for T=1 and T=20 and decreased for T=100. The percentiles of absolute values of difference in RMSD show decreased values compared to last year. No extreme errors are detected in 2024.

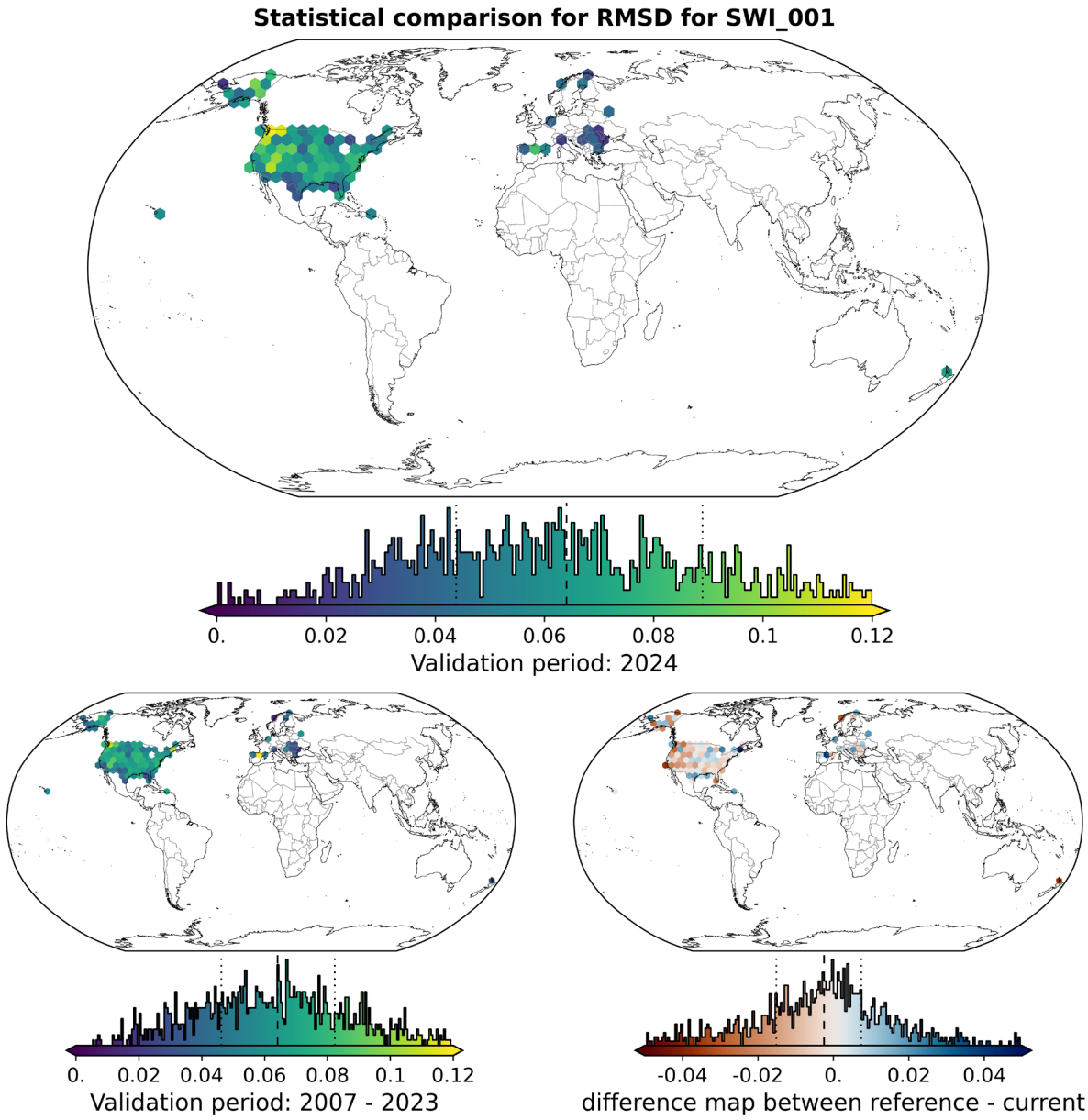


Figure 16: RMSD between SWI T=1 and in-situ data, as well the difference between the reference period (2007-01-01 until 2023-12-31) and the current validation period (2024-01-01 until 2024-12-31). In the difference map only the statistically significant differences are shown whereas the underlying distribution values show all data points. The distribution plots show the estimated kernel density plot as well as the median (dashed line) and the lower and upper quartile (dotted lines).

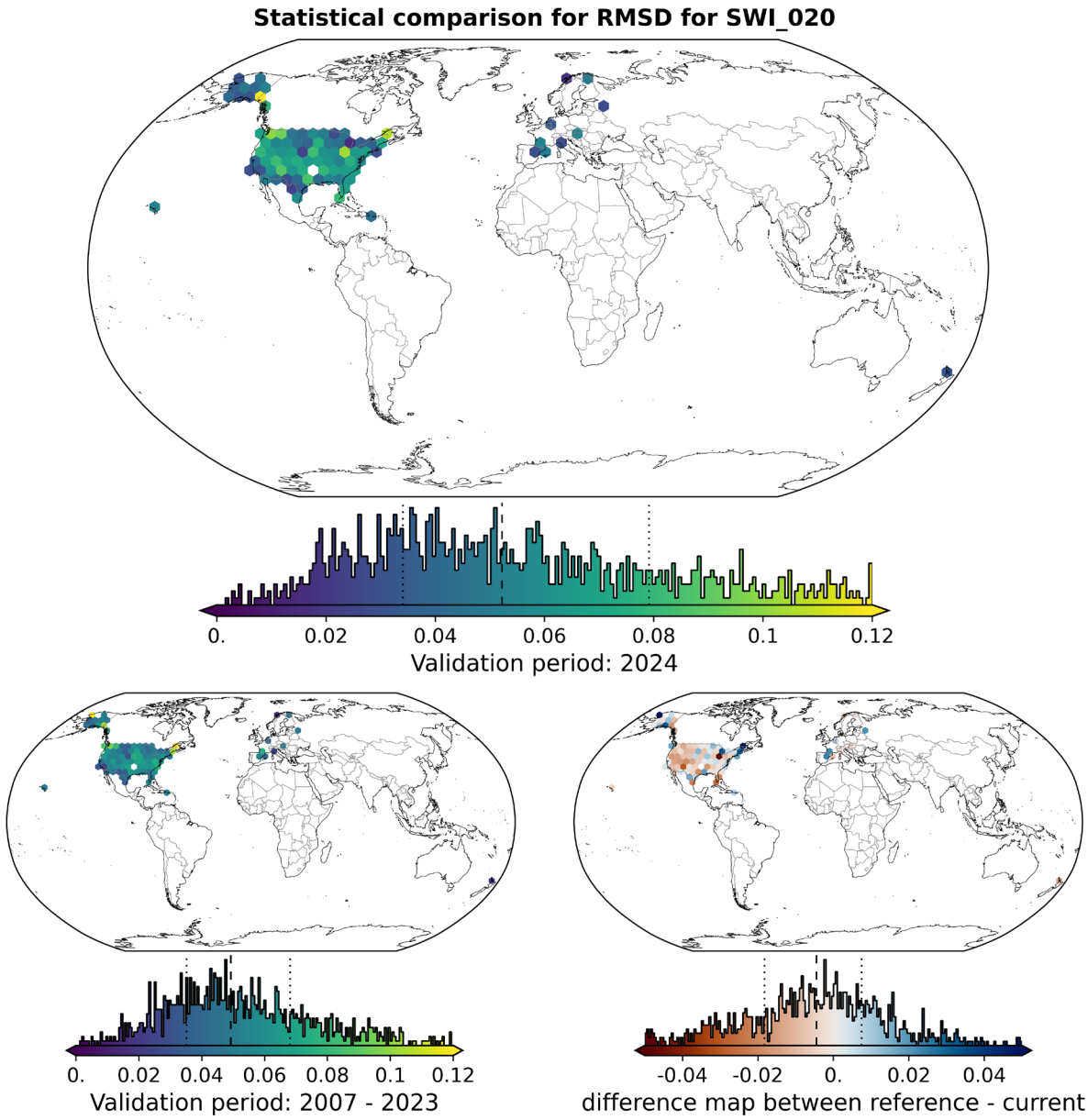


Figure 17: RMSD between SWI T=20 and in-situ data, as well the difference between the reference period (2007-01-01 until 2023-12-31) and the current validation period (2024-01-01 until 2024-12-31). In the difference map only the statistically significant differences are shown whereas the underlying distribution values show all data points. The distribution plots show the estimated kernel density plot as well as the median (dashed line) and the lower and upper quartile (dotted lines).

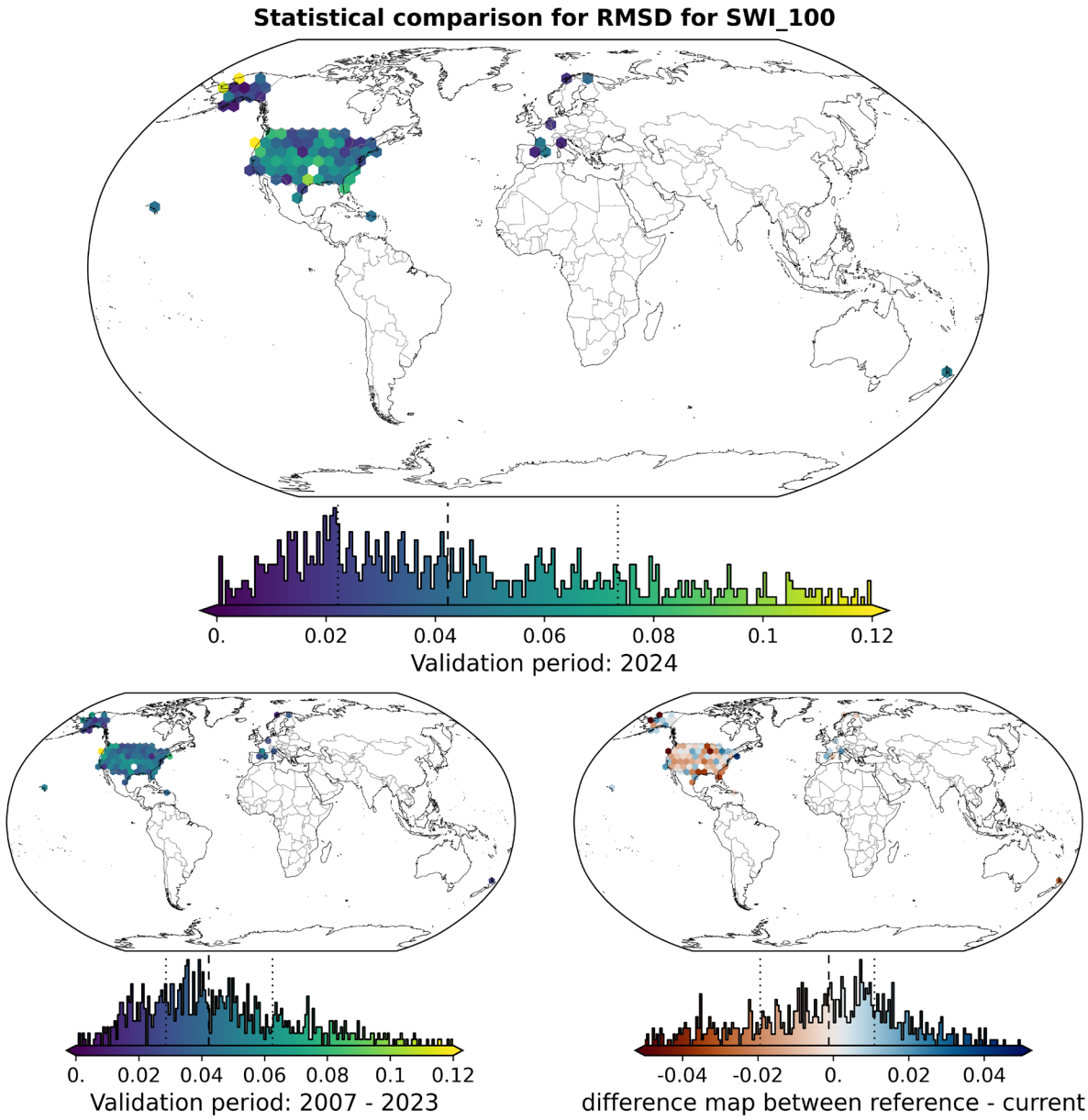


Figure 18: RMSD between SWI T=100 and in-situ data, as well the difference between the reference period (2007-01-01 until 2023-12-31) and the current validation period (2024-01-01 until 2024-12-31). In the difference map only the statistically significant differences are shown whereas the underlying distribution values show all data points. The distribution plots show the estimated kernel density plot as well as the median (dashed line) and the lower and upper quartile (dotted lines).

Table 11: Mean / median RMSD in SWI and in-situ observations as well as percentiles of the absolute values of the differences. All values in m^3/m^3.

	RMSD in current period		*absolute values of differences in RMSD*		
	mean	**median**	**50th percentile**	**70th percentile**	**80th percentile**
SWI T=1	0.071	0.064	0.012	0.019	0.027
SWI T=20	0.060	0.052	0.012	0.022	0.029
SWI T=100	0.054	0.042	0.014	0.027	0.033

5.1.2.2 Time Series for selected in-situ stations

In continuation of the analyses in previous VR reports, six in-situ stations were selected to be discussed in more detail. The selections take care that poor and good results are shown, and that different climate zones are represented. Station location and climate is listed in Table 12. For all in-situ data printed in Figure 19 to Figure 24, the first layer (measured at a depth from 0 to 0.05 m) is shown. For these plots, SWI with T=1 is plotted, which was found to be in general most corresponding to the 0 to 0.05m depth.

Table 12: Station details and climate classifications

Station Name	Network	latitude / longitude [°]	Köppen-Geiger Climate class
Ben Lomond peak	SNOTEL	41.38 N / 111.94 W	**Dsb**: cold – dry summer – Hot summer
Bushland #1	SCAN	35.17 N / 102.10 W	**Bsk:** semiarid – steppe – cold arid
Mercury 3 SSW	USCRN	36.57 N / 116.02 W	**Bwk:** arid – desert – cold arid
Selhausen	TERENO	50.86 N / 6.45 W	**Cfb:** warm temperate – fully humid – warm summer
Satu Mare	RSMN	47.72 N / 22.89 E	**Dfb:** cold – without dry season – Warm summer
SODAWS	FMI	67.36 N / 26.63 E	**Dfc:** cold – without dry season- cold summer

Table 13 shows values for Pearson rho and RMSD obtained at these stations. The station Mercury 3 SSW shows strong negative correlations, confirming the insight from last reports that there is a low agreement of SWI and in-situ data over arid areas. The poor agreement leads even to some "negative" SM values in the SWI signal, due to not applicable assumptions for the data scaling using the standard deviation method (Figure 21).

The station Bushland #1 is located in a semi-arid climate and shows medium agreement with SWIv3 data, and a closer look on the two time-series is of particular interest: while the general dynamics are agreeing well and the peaks coincide in time regularly, the level of highest and lowest soil moisture differ often, most notably in the years 2018, 2019, 2020, as well as 2023 and 2024 to a lesser extent. From this data, drying-out events appear more drastic in the SWIv3 time series than in the ground station observations (Figure 20).

A low correlation is observed over the FMI station SODAWS, located in northern Finland with cold climate and long data gap due to frozen soil. The low correlation is likely due to the challenging land cover type over which backscatter is likely more sensitive to temperature change or vegetation geometry than surface soil moisture (Figure 24).

The other European stations Satu Mare (Romania, Figure 23) and Selhausen (Germany, Figure 22) show high correlation as the SWIv3 product capture both the wet and dry events from the in-situ stations. Clear long-term seasonality, as well as short-term events, are well reproduced by the SWIv3 timeseries. Western Germany (where Selhausen station is located) experienced a higher-than-average wet year in 2024, and it is well captured by the SWIv3 product. In Romania, an underestimation of moisture content can be seen in SWIv3 product compared to in-situ dynamics, however, the performance remains high in 2024 as well.

Lastly, the Ben Lomond Peak station in Utah/USA shows a clear seasonality, with annual dry events during the summer period and wetting period in Autumn (winter period masked due to low temperature flag) (Figure 19). This alternation is reproduced by the SWIv3 data, but it often overestimates soil moisture during the drying periods. One interpretation is that the relatively large footprint of ASCAT might not capture such strong local dynamics.

Table 13: Pearson's correlation coefficient (rho) and Root mean square deviation (RMSD) for the selected in-situ stations. SWI T=5 was compared to in-situ soil moisture measured at a depth of 0.05m.

Station	rho	RMSD [m³/m³]
Ben Lomond peak	0.65	0.10
Bushland #1	0.48	0.08
Mercury 3 SSW	-0.74	0.04
Selhausen	0.67	0.06
Satu Mare	0.59	0.06
SODAWS	0.11	0.07

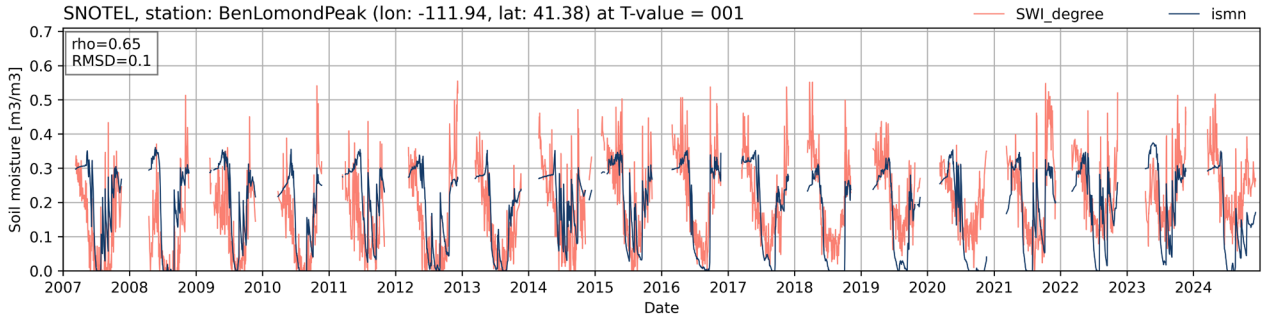


Figure 19: SWI and in-situ data for SNOTEL-station Ben Lomond Peak at depth=0.05m

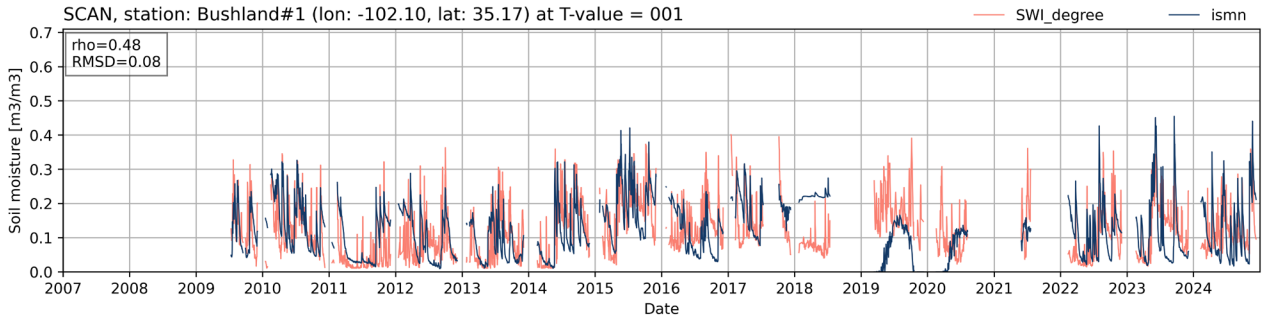


Figure 20: SWI and in-situ data for SCAN-station Bushland #1 at depth=0.05m

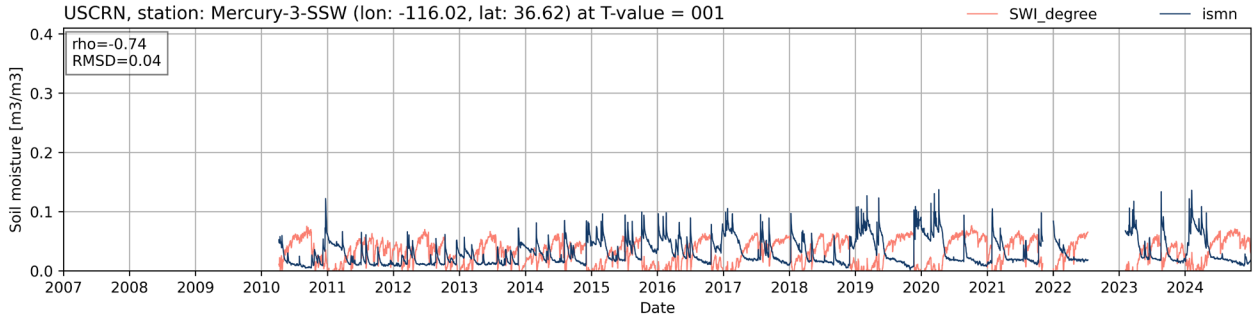


Figure 21: SWI and in-situ data for USCRN-station Mercury 3 SSW at depth=0.05m

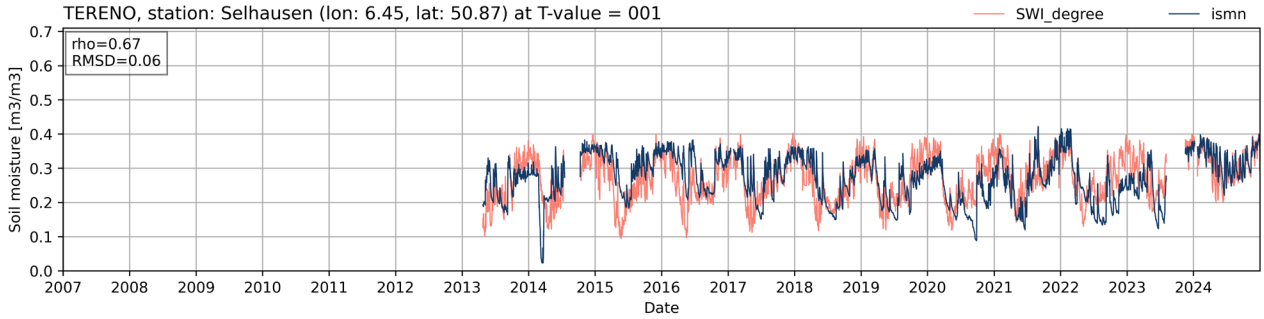


Figure 22: SWI and in-situ data for TERENO-station Selhausen at depth=0.05m

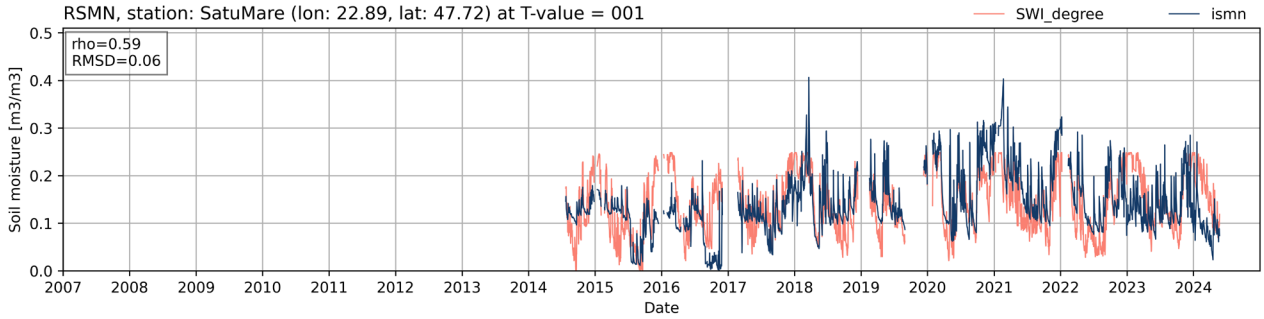


Figure 23: SWI and in-situ data for RSMN-station Satu Mare at depth = [0.00-0.05]m

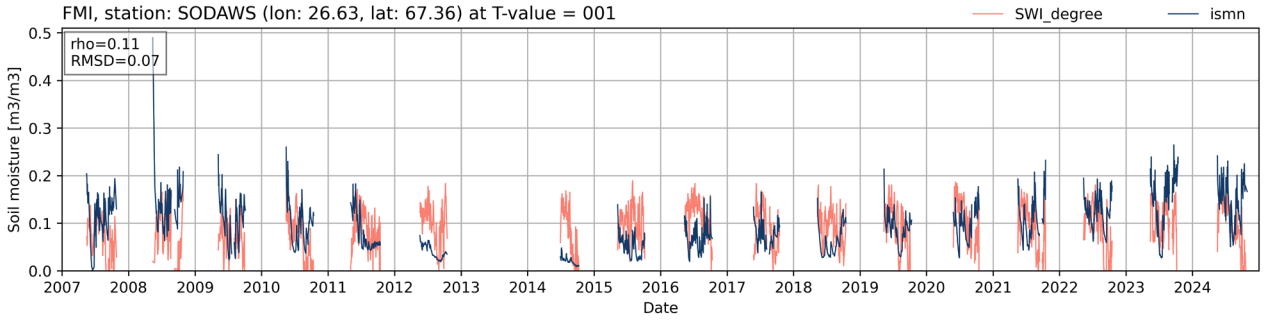


Figure 24: SWI and in-situ data for FMI-station SODAWS at depth=0.02m

5.2 REGIONAL ANALYSIS

The regional analyses for 2024 were carried out in three different continents, specifically in South America, Eastern Africa, and Central Europe, to illustrate how the CLMS SWI product performed in the case of recent drastic climate events leading to floods and droughts.

These comparisons are intended to provide a qualitative overview of the SWI, highlighting the link between soil water, precipitation, and vegetation indexes. The reference period for the SWI anomaly calculation is 17 years, from 2007 to 2023. Although relatively not long, 17 years of data is deemed sufficient to analyse SWI climatology and derive hydrological events such as droughts or floods.

5.2.1 Flooding in the south of Brazil

In May 2024, the south of Brazil (State of Rio Grande do Sul) experienced a record-breaking flood that affected over 2.4 million people and claimed 183 lives (Simoes-Sousa et al., 2025).

One of the worst-affected cities was the state's capital Porto Alegre (NF1), with a population of 1.4 million people. The waterway reached record levels of 5.33 meters on May 5th 2024. The city was cut from supply on water, electricity and food, and other 397 cities out of the state's 497 cities were declared into the state emergency (NF1).

Over the span of one week (from April 27th to May 2nd), the region received nearly a third of its expected yearly precipitation and many cities recorded between 500 and 700 mm of rain (NF2). Until the end of May, two more intense rain periods occurred in the region, ranging from 130 to 300 mm, and especially affecting the capital Porto Alegre (NF3, NF4).

The monthly anomaly calculation of SWI is displayed in Figure 25. It already shows positive anomalies of 20% in April. Those are also observed in May and, to a lesser extent, June. The temporal

and spatial dynamics of SWI anomalies match with the SPI values for the same time span, in Figure 26.

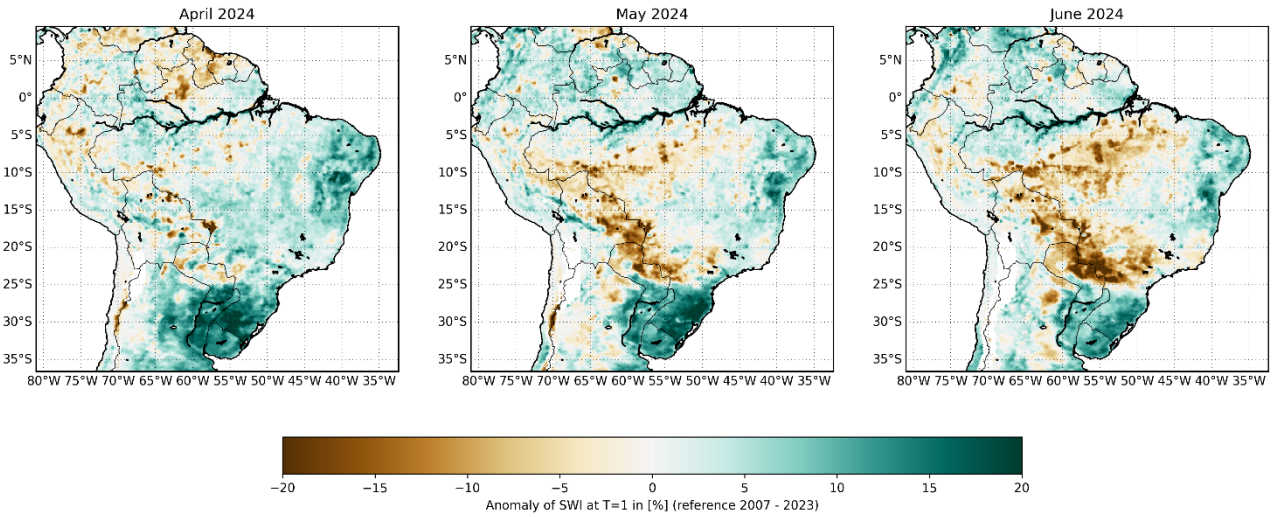


Figure 25: Anomalies of SWI at T=1 in (%) for South America during April to June, 2024

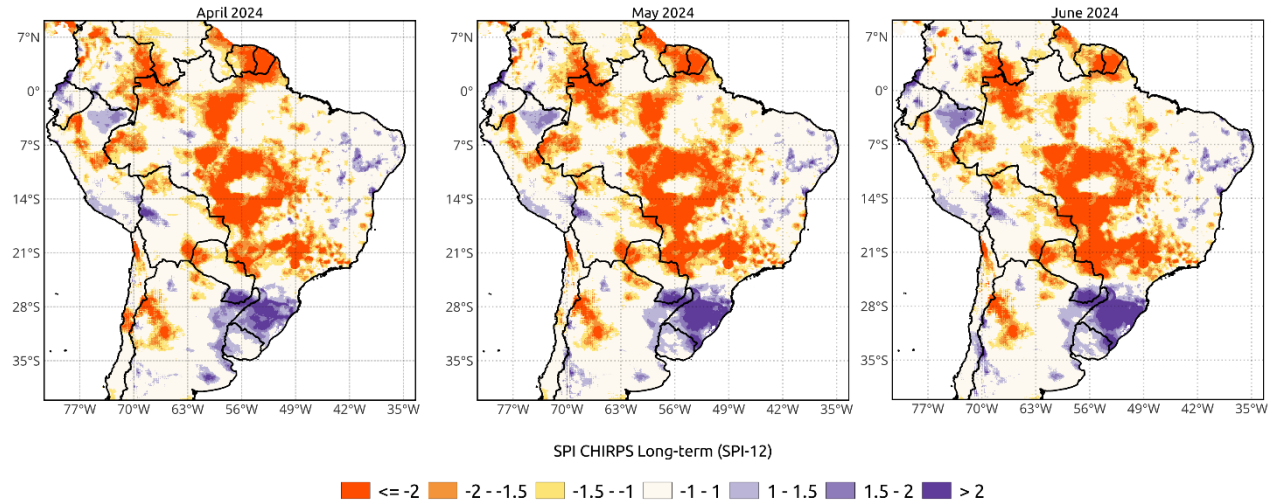


Figure 26: SPI CHIRPS Long-term (SPI-12) for South America during April to June, 2024

The Figure 27 shows anomalies for the period from 26 April 2024 until 6 May 2024 in South America categorized into four types: image (a) stands for 10-meter air temperature anomaly, (b) for

precipitable water vapor anomaly, (c) for anomalous height at 500-hPA and winds and (d) indicates mean precipitation. In images (a), (b) and (c), the anomalies are calculated based on 40-years' time range MERRA-2 data and image (d) in 24-years' time range of IMERG data. The area of the flood event is highlighted by a yellow outline (Simoes-Sousa et al., 2025). It can be summarised that region presented relevant anomaly for all four analyses, presenting considerably higher values for temperature, water vapor and precipitation than average and in the same area of the anomalies from SWI product, indicating that the abnormal rainfalls preceding the floods were captured by all the aforementioned variables.

The temperature anomaly in image (a) shows that in the south of the highlighted area there is a negative temperature anomaly, indicating the north-eastward propagation of an atmospheric cold front. The northern air mass of the prevailing heat wave may have blocked the cold front, resulting in the intense and localized precipitation in southern Brazil, as showed in image (d) (Simoes-Sousa et al., 2025).

Moreover, the heat was accompanied from notable low values from precipitable water vapor in the northeast area and the influence of the South American Low-Level Jet (indicated in the image (b) as SALLJ), which acts as a robust moisture corridor by transporting significant amounts of water vapor from the Amazon region to the south of the country. The image (c) was implemented to corroborate with the investigation of atmospheric blocking that could have resulted in the intense rains. The red polygon was identified as an area of atmospheric blocking by the study of Simoes-Souza et al. (2025). Those events are generally associated with extreme drought conditions in some locations and heavy precipitation in others, which resulted in extreme heat and drought in central Brazil and intense rainfall in the south (Simoes-Sousa et al., 2025). The drought and heat are also identified in the SWI images by the presence of big areas with negative anomaly values for May and June (Figure 25).

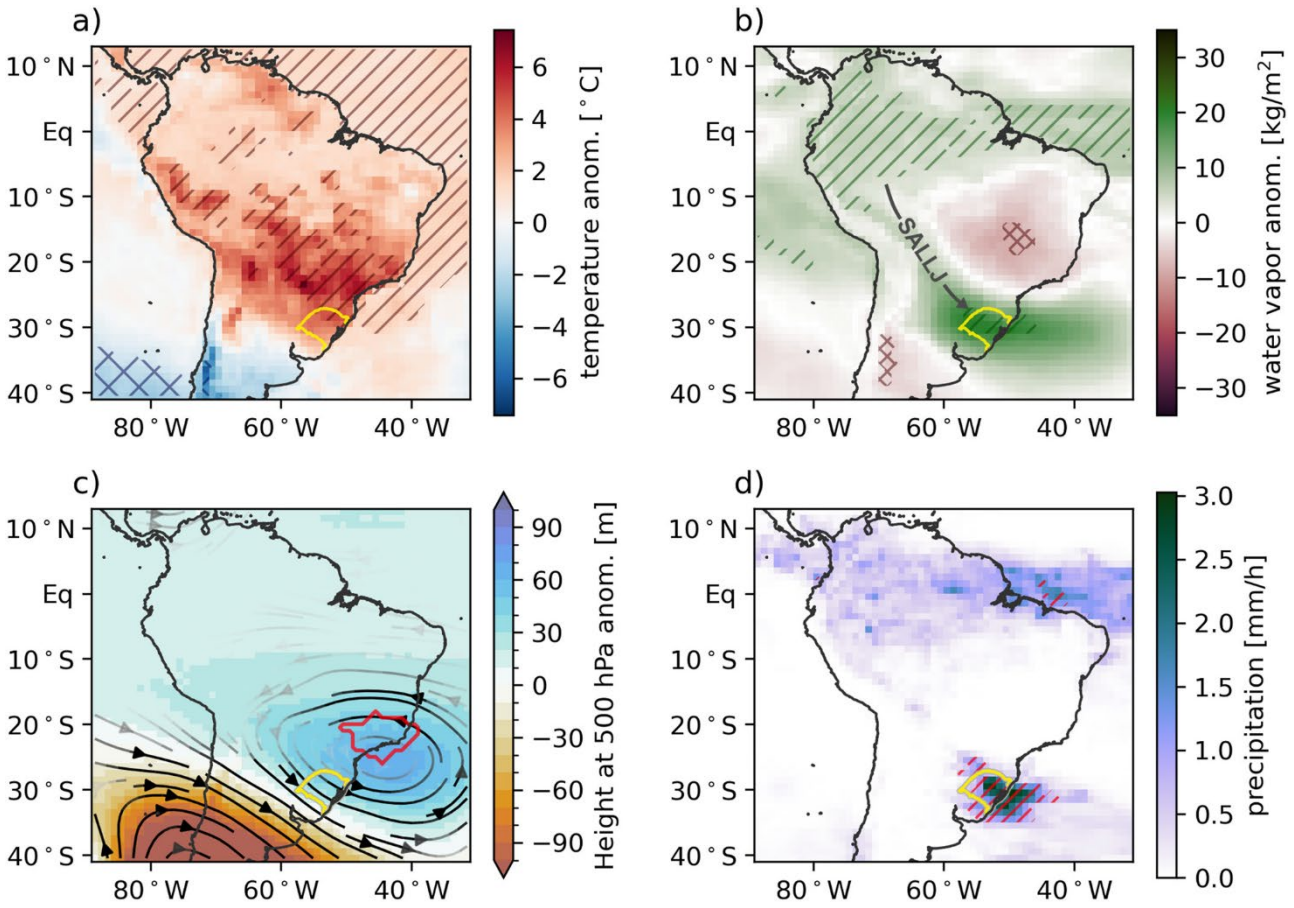


Figure 27: (a) 10-m air temperature anomaly, (b) precipitable water vapor anomaly, (c) anomalous height at 500-hPa and winds, and (d) mean precipitation. The state Rio Grande do Sul is highlighted in yellow. the maps are averages for the period 26 April to 6 May 2024. (a-c) uses as comparison for anomaly of 40-yearstime range from MERRA-2 and (d) uses 24 years' time range from IMERG. The grey arrow in panel b represents the South American Low-Level Jet (SALLJ). The red polygon in panel c depicts the region of atmospheric blocking. Diagonal hatching indicates values above the 90th percentile, and cross-hatching indicates values below the 20th percentile. (Simoes-Sousa et al., 2025)

5.2.2 Kenya-Tanzania floods

In 2024, the rainy season in Eastern Africa (March to May) was drastically intensified by the El Niño phenomenon. In Kenya, the United Nations estimated that, as of May 16th, extreme rainfall causing flooding events led to 291 deaths, 188 injuries and 75 missing persons (OCHA, 2024a). Over 270.000 people were displaced and almost more than 400.000 were overall affected by persistent heavy rains

and flooding. Moreover, the same report informed that around 11.000 livestock was lost and approximately 190 square kilometres of cropland were destroyed (OCHA, 2024b).

In Tanzania, 210.000 people have been affected by floods, 250 people have been injured, and 161 people have lost their lives. Over 800 km² of farmland have been affected. Other countries hit by the event were Somalia, with more than 160.000 people impacted, and Burundi with 22.000 affected (UNICEF, 2024).

Anomalies of SWI are displayed in Figure 28. SPI is showcased in Figure 29, and the Kenya monthly rainfall anomalies from the Kenya Meteorological Department (Nying'uro, 2024) are displayed in Figure 30. Whereas media footage reported flood onsets in April/May, high SWI and SPI anomalies were already detected from March on, and anomalies of around 20% are observed. The monthly rainfall anomalies indicate a slight increase in March and a significant surge in April. When put into respect with the SPI values, SWI anomalies provided an early and spatially consistent indication of the upcoming flooding event in this use case.

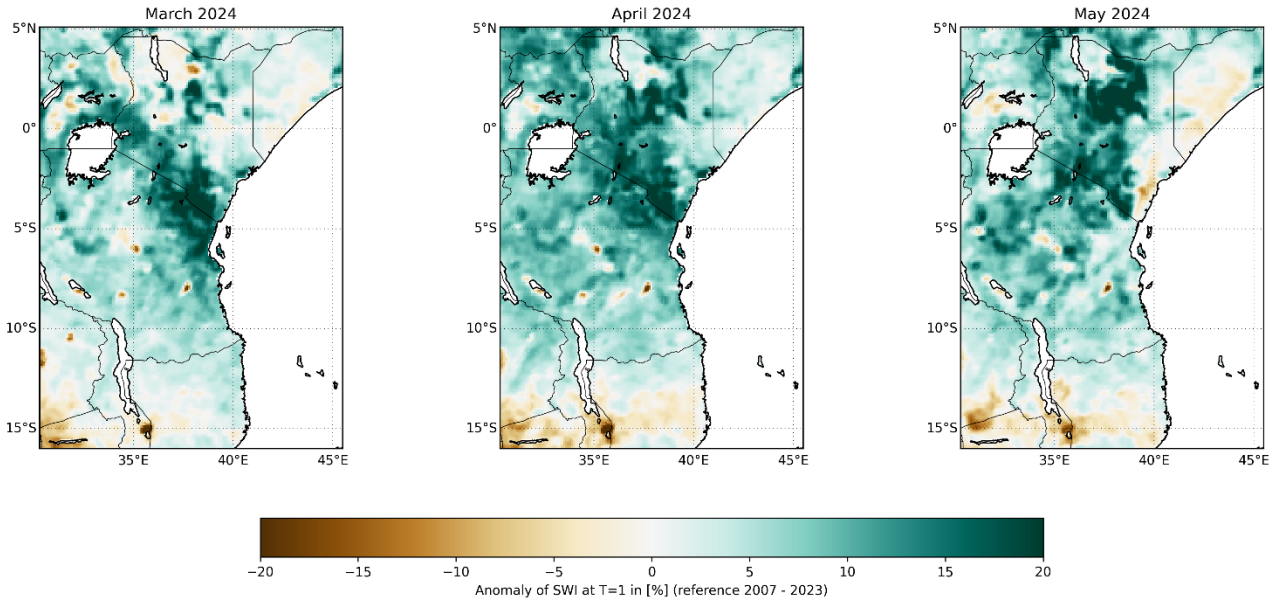


Figure 28: Anomalies of SWI at T=1 in (%) for eastern Africa during March to May, 2024

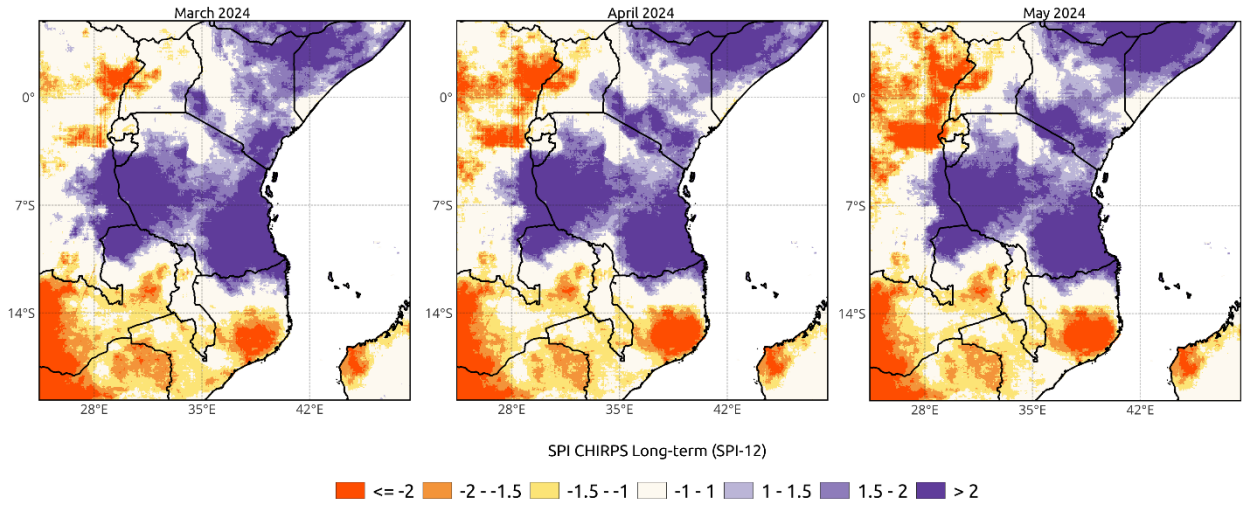


Figure 29: SPI CHIRPS Long-term (SPI-12) for eastern Africa during March to May, 2024

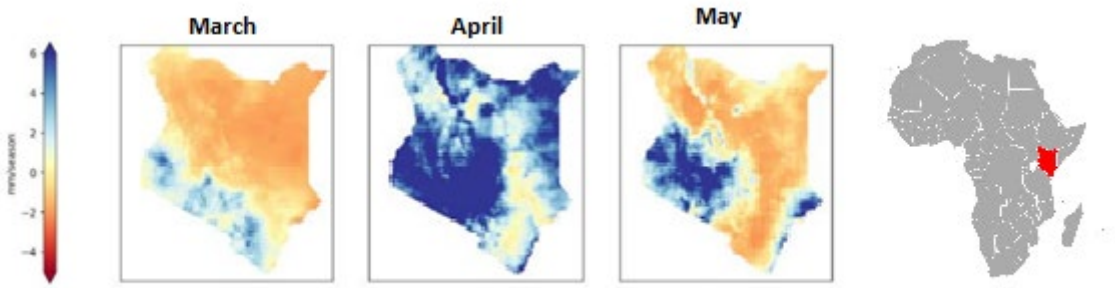


Figure 30: Kenya monthly rainfall anomalies from the Kenya Meteorological Department (Nying'uro, 2024)

5.2.3 Hungary and Serbia drought

Hungary and northern Serbia faced extremely high temperatures in the summer of 2024. It was the hottest year ever recorded in more than 130 years of measurements (NF5). Furthermore, the area faced a long period of time with temperatures exceeding 30ºC and not falling below 20ºC overnight (NF5).

Additionally, the media reported that farmers declared drought damage on more than 300.000 ha of land by the end of August. The prolonged drought conditions and high temperatures led to at least

30% of withered crops (NF6). At the beginning of September, Hungary's Minister for Agriculture informed that maize plantation was severely affected on 235.000 hectares, which corresponds to more than 25% of all planted areas in the country (NF6). The national average monthly precipitation total was 24.7 mm for August, which accounts for only 42% of the expectation from the multi-year average for 1991-2020 (HungaroMet, 2025).

In Serbia, the harvesting period started a month earlier, and yields for sunflower, soybean, and fruits saw a decrease of around 25 to 30%, generating an economic loss estimation of around half a billion Euros (NF7). The SWI product displayed in Figure 31 shows accentuated negative values for the region of southern Hungary and northern Serbia. The SPI-12 values shown in Figure 33 highlight extreme precipitation anomalies in July and August. Both SWI and SPI detect the peak of the drought for the month of August, matching the media report on the state of the climate.

When analyzing the fAPAR for the region (Figure 32), a strong negative anomaly in August is observed, concentrated in the region of Southern and central Hungary and northern Serbia. These results match with SPI and SWI spatial dynamics, but with a temporal delay of one month. This is expected as the vegetation response to precipitation and soil moisture deficit is expected to be delayed. Plant water stress takes time to impact physiological processes and ultimately the monitored absorbed photosynthetic radiation (Copernicus Global Drought Observatory, 2023).

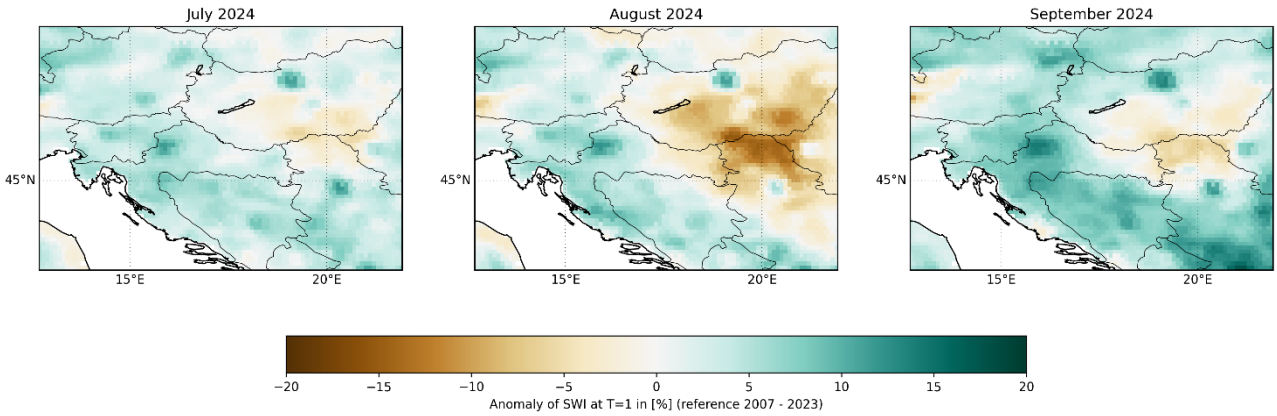


Figure 31: Anomalies of SWI at T=1 in (%) for Hungary and surroundings during July to September, 2024

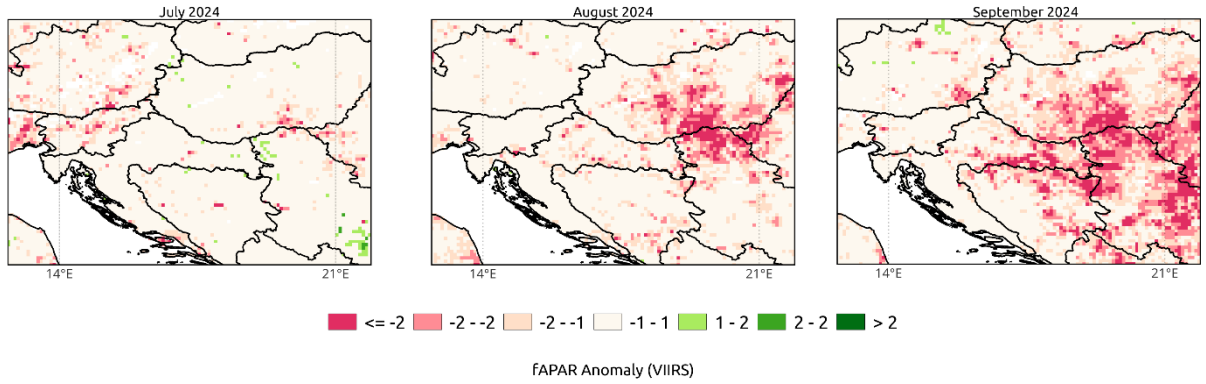


Figure 32: fAPAR Anomaly (VIIRS) for Hungary and surroundings for the first day of the months July to September, 2024

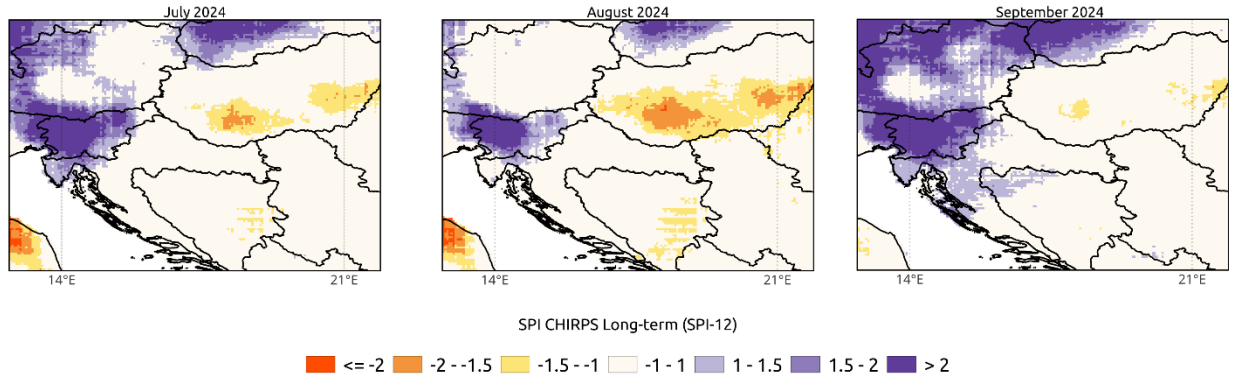


Figure 33: SPI CHIRPS Long-term (SPI-12) for Hungary and surroundings during July to September, 2024

6 CONCLUSIONS

6.1 GLOBAL ANALYSIS

The SWI and SSF were compared to the output of a land surface model as well as in-situ data. The model was GLDAS-Noah V2.1, whereas the in-situ data was taken from suitable networks of the ISMN. The SWI was compared to different soil moisture layers of these reference datasets (Table 6), whereas the SSF was assessed using the model's first soil temperature layer. Overall, no or only little deviations between SWI data from the reference period (2007-2023) and the year 2024 were found, showing overall a weak performance over tropical rainforests, subarctic areas, and deserts, and a good performance over Europe, subtropical Africa, India, Central and East Asia, non-arid Australia and the (non-polar) Americas.

The matrix in Table 14 shows in summary the compliance with the CGLOPS requirements (Section 2.1) for the Soil Water Index (SWI).

Table 14: Compliance matrix for the comparisons against GLDAS Noah model and ISMN in-situ reference data. Plotted values are percentages of grid points that fulfil the CGLOPS requirements for Threshold and Goal (see Section 2.1). The reference period (2007-01-01 until 2023-12-31) is denoted "ref", and the current validation period (2024-01-01 until 2024-12-31) is denoted "cur". Results for the current validation period and the goal requirement are highlighted in bold

Variable	CGLOPS Threshold				CGLOPS Goal			
	SWI 0.10 m^3/m^3				**SWI 0.04 m^3/m^3**			
	GLDAS		ISMN		GLDAS		ISMN	
[%]	*ref*	***cur***	*ref*	***cur***	*ref*	***cur***	*ref*	***cur***
SWI T=1	*99*	***98***	*82*	***89***	*38*	***34***	*18*	***21***
SWI T=20	*99*	***97***	*86*	***94***	*31*	***27***	*33*	***34***
SWI T=100	*98*	***93***	*86*	***95***	*43*	***39***	*45*	***47***

For GLDAS and the current period (i.e. 2024), at the different soil depth layers, between 93% and 98% of all locations meet the threshold of 0.10 m^3/m^3, whereas between 27% and 39% of the grid points meet the goal of 0.04 m^3/m^3. Compared to the previous year (Table 1) values remain stable, but there is a small decrease for T=1 and T=20 for the 0.04 m^3/m^3 goal, and for T=100 for the 0.10 m^3/m^3 goal.

Overall, the comparisons against ISMN show similar compliance compared to 2023 (Table 1) with an increase of compliance for T=20 and T=100. For the current period (i.e. 2024), at the different soil depth layers, between 89% and 95% of all locations meet the threshold of 0.10 m^3/m^3, whereas between 21% and 47% of the grid points meet the goal of 0.04 m^3/m^3.

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6.2 REGIONAL ANALYSIS

The regional analysis focused on different continents in 2024, analysing SWI values over South America, Eastern Africa, and Central Europe, and comparing them with auxiliary products and regional climatic news.

The SWI product provided spatially detailed overview of the 2024 floods in South America, and the spatial patterns corresponds to hydrology- or vegetation-products, such as SPI. Moreover, the meteorologically-derived metrics from the study of Simoes-Souza et al. (2025) (10-meter air temperature anomaly, precipitable water vapor anomaly, anomalous height at 500-hPa and winds, and mean precipitation) are well-aligned with the results presented by the SWI product.

Furthermore, over Eastern Africa, SWI anomalies matched with SPI temporal variability and agree on spatial patterns of heavy rainfall events. The rainy season was characterised by some extremely wet months, which led to a vegetation surge as well as large flood events. These events spatially correspond well with SWI positive anomalies in 2024.

Representing exceptional dry and hot conditions in Central Europe, the rather severe drought in Hungary and Serbia is captured too by the SWI anomalies. It corresponds well with the spatial patterns of fAPAR, as vegetation appear to be stressed by insufficient precipitation inputs and ongoing water storage depletion, as well as the SPI that shows intense anomaly in the month of August in Hungary.

In summary, the anomalies calculated from the SWI product suggest that the SWIv3 shows good skill in indicating regional patterns of outstanding hydrological events. This skill was found to apply in both cases: flood and drought events, i.e. soil water surpluses and deficits.

7 RECOMMENDATIONS AND PERSPECTIVES

The overall performance of the CLMS SWIv3 and SSF products is consistent with the previous years' scientific quality monitoring. No inconsistencies in the SWI were detected in this year's VR analysis. The detected differences between the reference period and the current validation period were not consistently in one direction, as already known from the past year-to-year fluctuations.

The regional analyses offer interesting insights in medium- and large-scaled hydrological events and should be continued in coming VR exercises, as they not only document the skill of the SWI product to reflect droughts and rainfall anomalies (leading i.e. to floods), but also help to improve the understanding of spatio-temporal dynamics of such processes.

As the product quality is stable during the last year, we recommend a continuation of using the SWIv3 product. In general, the products appear to have high fidelity in the shallower soil layers, concerning the timing of peaks and lows. The known poor performance over arid regions persists in 2024, but will be expected to be improved through the currently implemented product updates (e.g. the upstream developments in H SAF ASCAT soil moisture towards the sub-surface scatter modelling).

Several algorithmic enhancements developed in dedicated third party research activities are currently in the process of being implemented in the CLMS operational production chains for soil moisture. The improvements for the SWIv3 will be centred on the upgrades of the input EUMETSAT H SAF SSM product, featuring a higher spatial resolution of this input, an improved vegetation characterisation, a masking of ill-posed subsurface scattering areas, and a dynamic parameterization of the dry- and wet-references to correct for slow-acting trends coming from long-term land cover changes. Furthermore, with the upcoming H SAF upgrades, the Surface State Flag will undergo a refinement, taking up the newly available forecast data for frozen soil and snow cover. The CLMS users can therefore expect improved quality and reliability of the future SWI product.

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